



Seismological Society of America

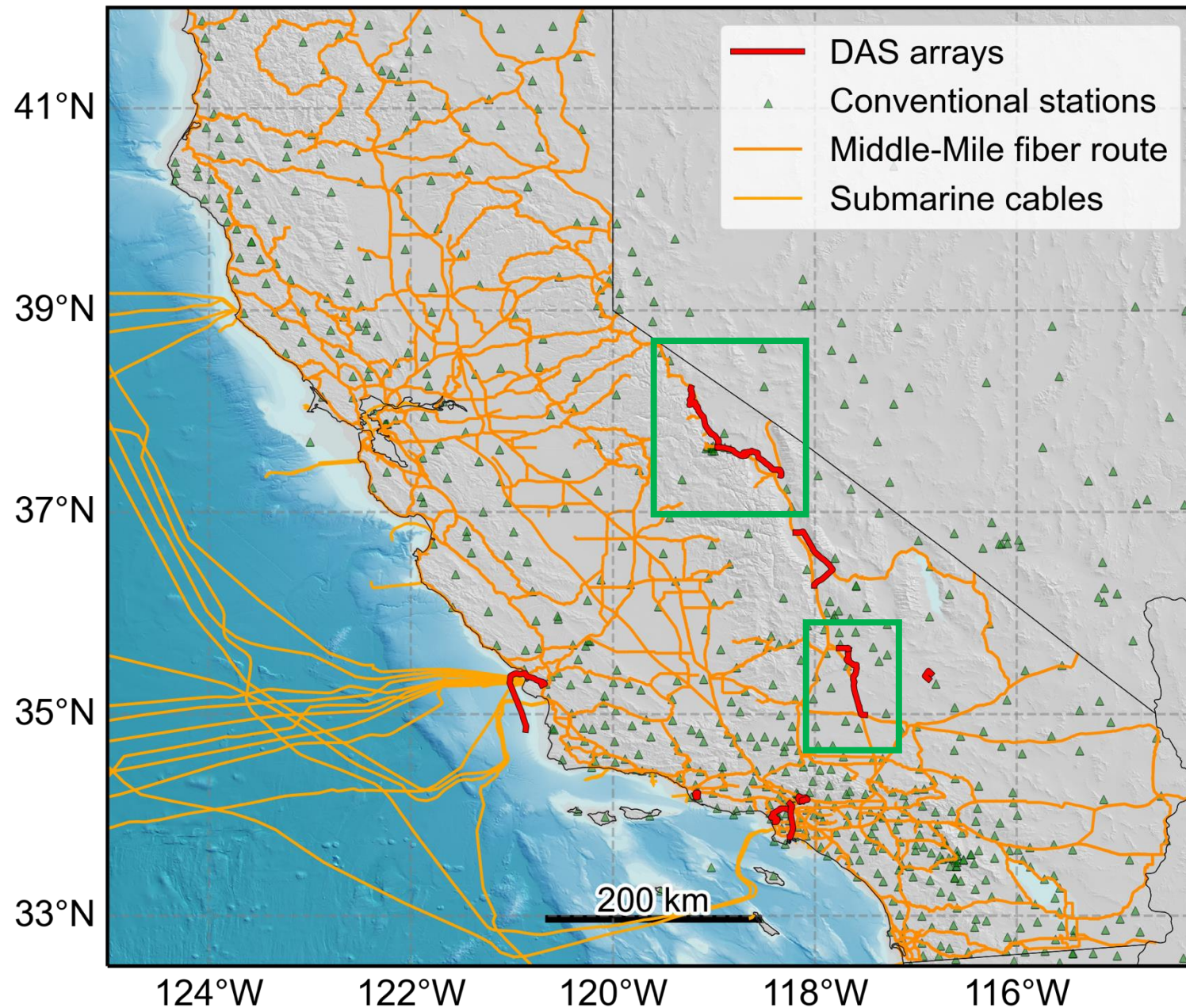


SSA2026 Workshop
Introduction to DAS for Seismology:
From Data Acquisition to Analysis

Earthquake Focal Mechanisms with DAS

Ettore Biondi, Jiaxuan Li*, Tianfan Yan, Chun Zhang, Weiqiang Zhu (alphabetical order)

Big data, Big challenge, New opportunities



- Use telecom cable
- To the ocean bottom
- Real-time telemetry
- High spatial resolution

Produces big data:

- ~40,000 recording channels
- ~600 GB/day
- ~200 TB/year
- NCEDC+SCEDC: ~2000 stations @ 100 Hz

Challenges:

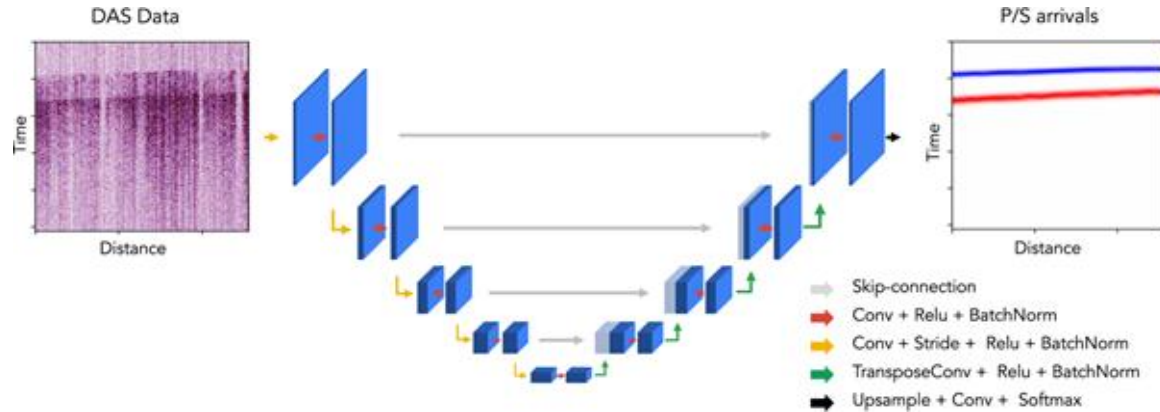
- Dense array
- Noisier than broadband stations
- Single-component sensing

Need:

- New methods/algorithms
- New computing paradigms
- ...

Earthquake detection, phase picking, magnitude

Detection and phase picking



(Zhu et al., 2023)

Magnitude estimation

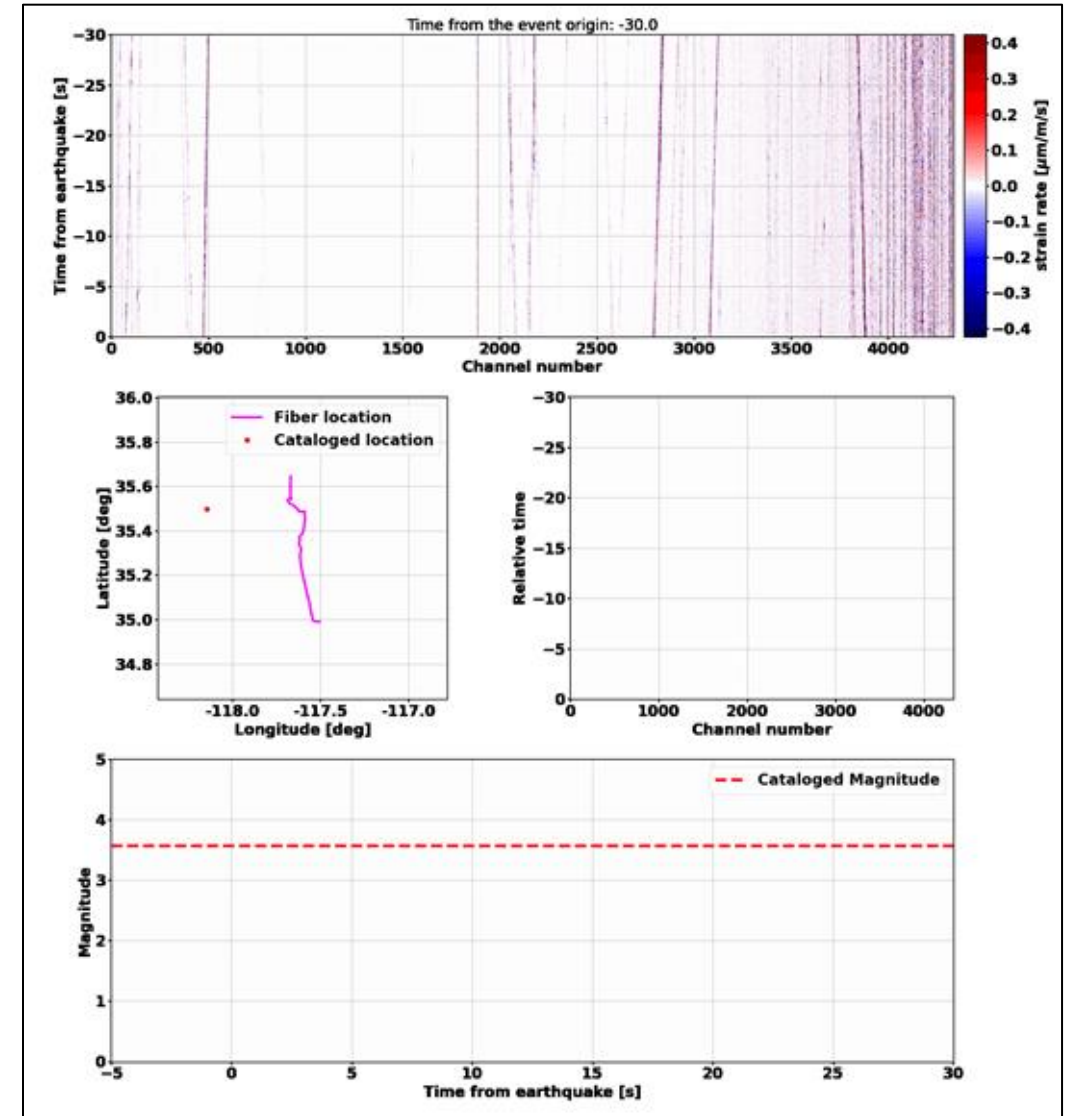
Scaling relation:

$$\log_{10} E_i = aM + b \log_{10} D_i + K_i$$

Peak strain rate Catalog magnitude Hypocentral distance Site term

(Yin et al., 2023)

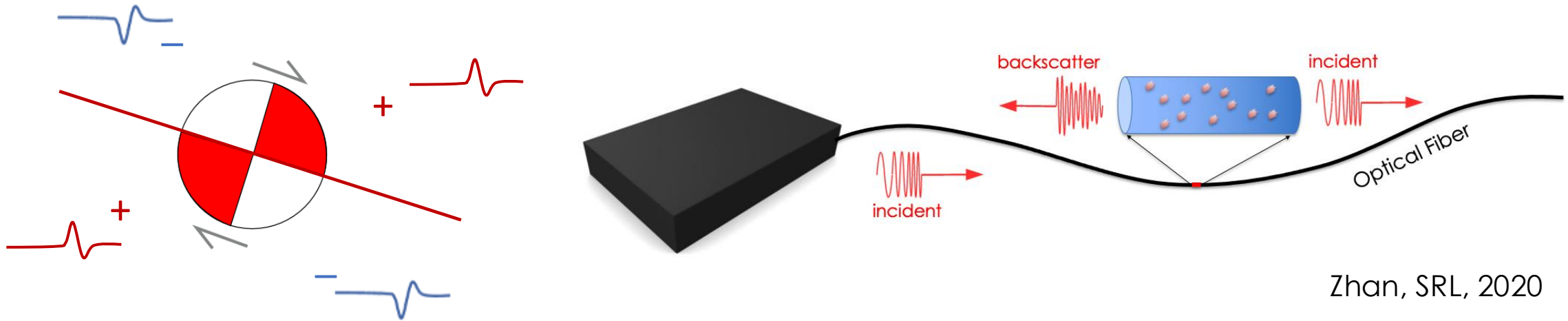
Real-time processing



(Biondi et al., 2026)

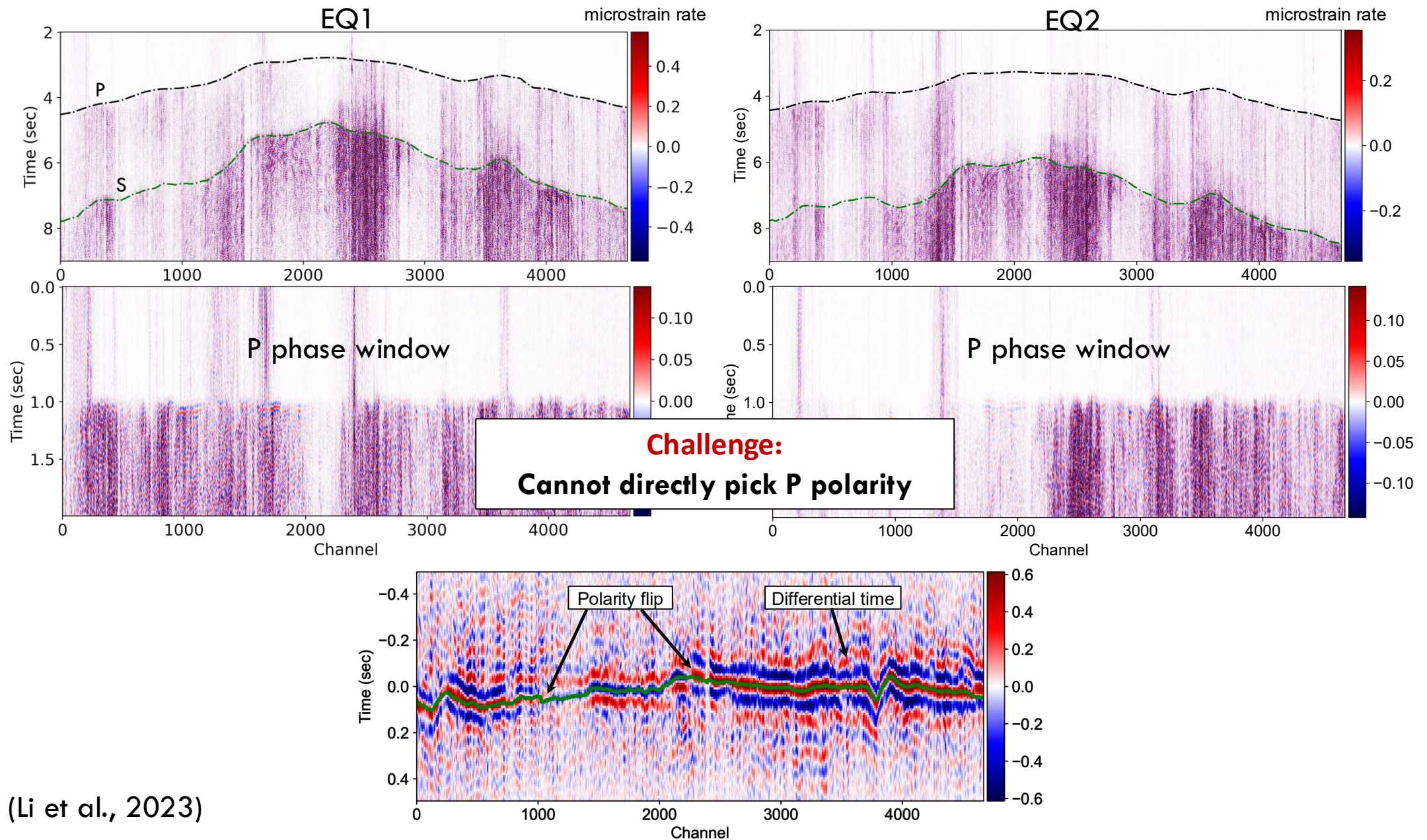
Earthquake focal mechanisms with DAS

- Focal mechanisms of earthquakes provide critical in-situ information about fault/fracture orientation and stress status/rotation.



- Distributed acoustic sensing provides a relatively new dataset for investigating focal mechanisms.
 - **Long sensing range:** Fill the angle gap for focal mechanism inversion.
 - **High spatial resolution:** Put super tight constraints on focal plane orientations.
 - Challenging due to the single-component sensing and low SNR.

First-arrival polarity information on DAS



(Li et al., 2023)

Invert for Absolute Polarities from Relative Measurements

Methods:

unknowns

Polarity vector at station k:

$$\vec{p}_k = \begin{bmatrix} P_{1k} \\ P_{2k} \\ \vdots \\ P_{Nk} \end{bmatrix}$$

N events K stations

IF: known polarities:

$$\vec{q}_k = \begin{bmatrix} Q_{1k} \\ Q_{2k} \\ \vdots \\ Q_{Mk} \end{bmatrix}$$

M events K stations

measurements

Relative polarity matrix at station k:

$$\mathbf{R} = \begin{bmatrix} P_{1k}Q_{1k} & P_{1k}Q_{2k} & \cdots & P_{1k}Q_{Mk} \\ P_{2k}Q_{1k} & P_{2k}Q_{2k} & \cdots & P_{2k}Q_{Mk} \\ \vdots & \vdots & \ddots & \vdots \\ P_{Nk}Q_{1k} & P_{Nk}Q_{2k} & \cdots & P_{Nk}Q_{Mk} \end{bmatrix}_{N \times M}$$

Templates with known polarities

First column vector of $\mathbf{U} \rightarrow$ Polarity P_k

$$\text{SVD: } \mathbf{R}^{<kk>} = \mathbf{U}^{<kk>} \mathbf{S}^{<kk>} \mathbf{V}^{<kk>*}$$

(Shelly et al., 2016)

Invert for Absolute Polarities from Relative Measurements

Methods:

$$\mathbf{R} = \begin{bmatrix} P_{1k}Q_{1k} & P_{1k}Q_{2k} & \cdots & P_{1k}Q_{Mk} \\ P_{2k}Q_{1k} & P_{2k}Q_{2k} & \cdots & P_{2k}Q_{Mk} \\ \vdots & \vdots & & \vdots \\ P_{Nk}Q_{1k} & P_{Nk}Q_{2k} & \cdots & P_{Nk}Q_{Mk} \end{bmatrix}_{N \times M}$$

For **DAS**: **Unknown...**

$$\mathbf{R}_{\langle kl \rangle} = \begin{bmatrix} P_{1k}P_{1l} & P_{1k}P_{2l} & \cdots & P_{1k}P_{Nl} \\ P_{2k}P_{1l} & P_{2k}P_{2l} & \cdots & P_{2k}P_{Nl} \\ \vdots & \vdots & & \vdots \\ P_{Nk}P_{1l} & P_{Nk}P_{2l} & \cdots & P_{Nk}P_{Nl} \end{bmatrix}_{N \times N}$$

$$\text{SVD: } \mathbf{R}^{\langle kk \rangle} = \mathbf{U}^{\langle kk \rangle} \mathbf{S}^{\langle kk \rangle} \mathbf{V}^{\langle kk \rangle *}$$

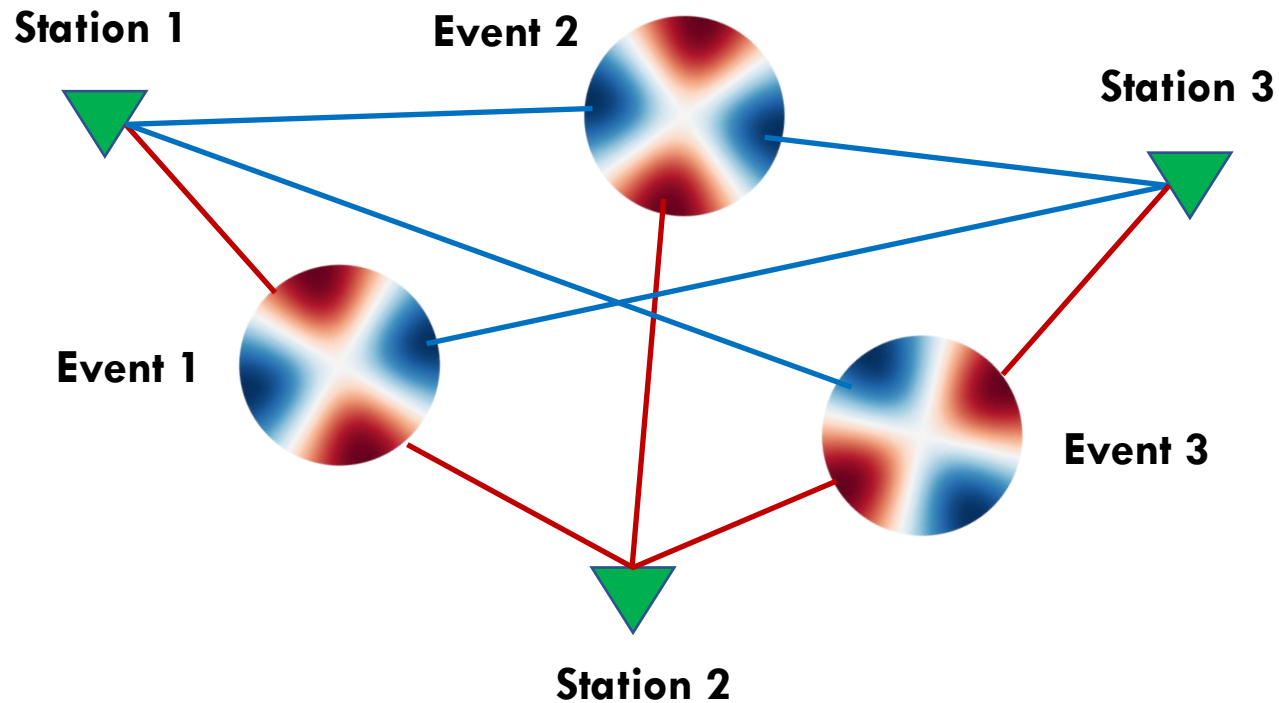
Problem of “-1” ambiguity!

We could include cross-correlations between different events at different channels to resolve the “-1” ambiguity.

Sign correction: N=M=3

Special case:

When $N=3$ and $M=3$:



Polarity vectors: $\tilde{\mathbf{p}}_k = \begin{bmatrix} P_{1k} \\ P_{2k} \\ \vdots \\ P_{Nk} \end{bmatrix}$

		Stations:		
		p_1	p_2	p_3
Events	1	$\begin{bmatrix} 1 \end{bmatrix}$	$\begin{bmatrix} 1 \end{bmatrix}$	$\begin{bmatrix} -1 \end{bmatrix}$
	2	$\begin{bmatrix} -1 \end{bmatrix}$	$\begin{bmatrix} 1 \end{bmatrix}$	$\begin{bmatrix} -1 \end{bmatrix}$
	3	$\begin{bmatrix} -1 \end{bmatrix}$	$\begin{bmatrix} 1 \end{bmatrix}$	$\begin{bmatrix} 1 \end{bmatrix}$

Sign correction: N=M=3

Relative polarity matrix

$$\mathbf{R}^{<kk>} = \begin{bmatrix} P_{1k}P_{1k} & P_{1k}P_{2k} & \cdots & P_{1k}P_{Nk} \\ P_{2k}P_{1k} & P_{2k}P_{2k} & \cdots & P_{2k}P_{Nk} \\ \vdots & \vdots & & \vdots \\ P_{Nk}P_{1k} & P_{Nk}P_{2k} & \cdots & P_{Nk}P_{Nk} \end{bmatrix}$$

At station k=1:

$$\mathbf{R}^{<1,1>} = \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}^T \otimes \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix} = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 1 \\ -1 & 1 & 1 \end{bmatrix}$$

Polarity vector from SVD
with a “-1” ambiguity at each station

$$\mathbf{R}^{<1,1>} = \mathbf{U}\mathbf{S}\mathbf{V}$$

Singular values: $\mathbf{S} = \text{diag}(3, 0, 0)$

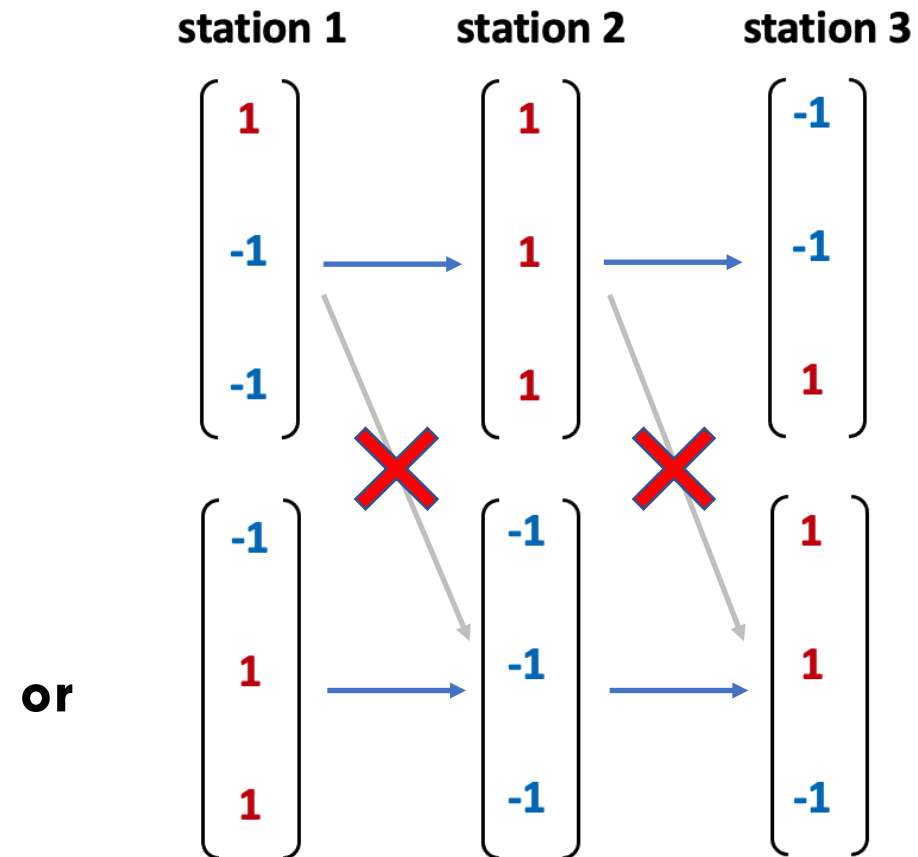
First column vector of $\mathbf{U}$:

$$\frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix} \quad \text{or} \quad \frac{1}{\sqrt{3}} \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$$

Ambiguity of -1

Sign correction: N=M=3

Correct the “-1” ambiguity using inter-station cross correlations



Real Polarity vectors

		Stations:		
		p_1	p_2	p_3
Events	1	$\begin{bmatrix} 1 \end{bmatrix}$	$\begin{bmatrix} 1 \end{bmatrix}$	$\begin{bmatrix} -1 \end{bmatrix}$
	2	$\begin{bmatrix} -1 \end{bmatrix}$	$\begin{bmatrix} 1 \end{bmatrix}$	$\begin{bmatrix} -1 \end{bmatrix}$
	3	$\begin{bmatrix} -1 \end{bmatrix}$	$\begin{bmatrix} 1 \end{bmatrix}$	$\begin{bmatrix} 1 \end{bmatrix}$

Sign correction: general case

Relative polarity matrix at station k : $\mathbf{R}^{<kk>} = \begin{bmatrix} P_{1k}P_{1k} & P_{1k}P_{2k} & \cdots & P_{1k}P_{Nk} \\ P_{2k}P_{1k} & P_{2k}P_{2k} & \cdots & P_{2k}P_{Nk} \\ \vdots & \vdots & & \vdots \\ P_{Nk}P_{1k} & P_{Nk}P_{2k} & \cdots & P_{Nk}P_{Nk} \end{bmatrix}$

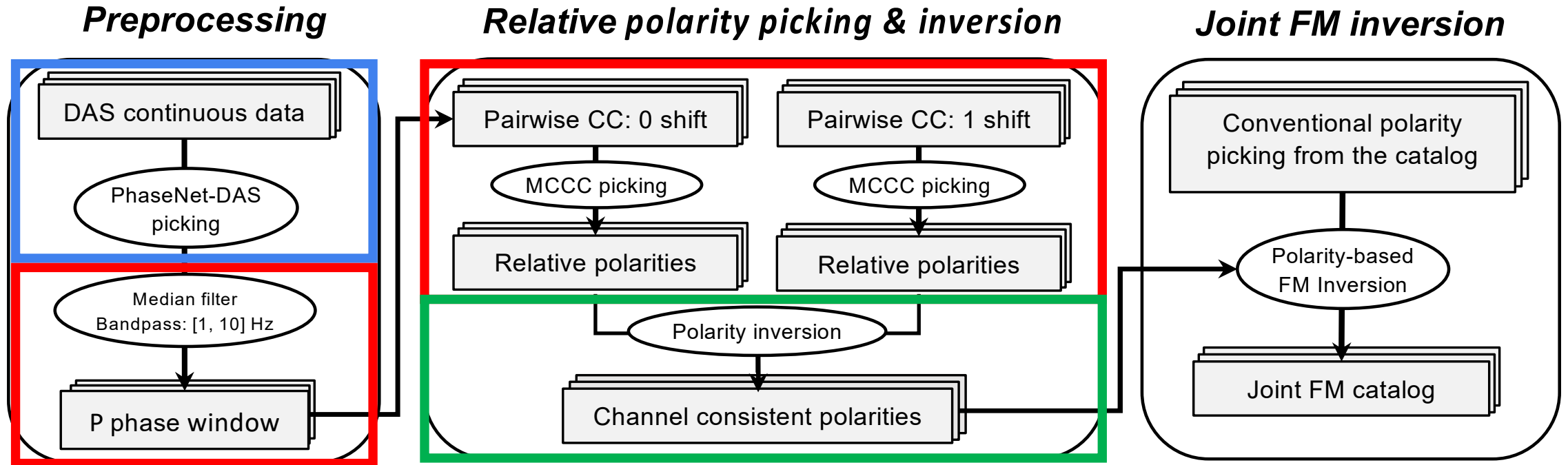
Outer product of polarity vector at station k :

SVD: $\mathbf{R}^{<kk>} = \mathbf{U}^{<kk>} \mathbf{S}^{<kk>} \mathbf{V}^{<kk>*}$

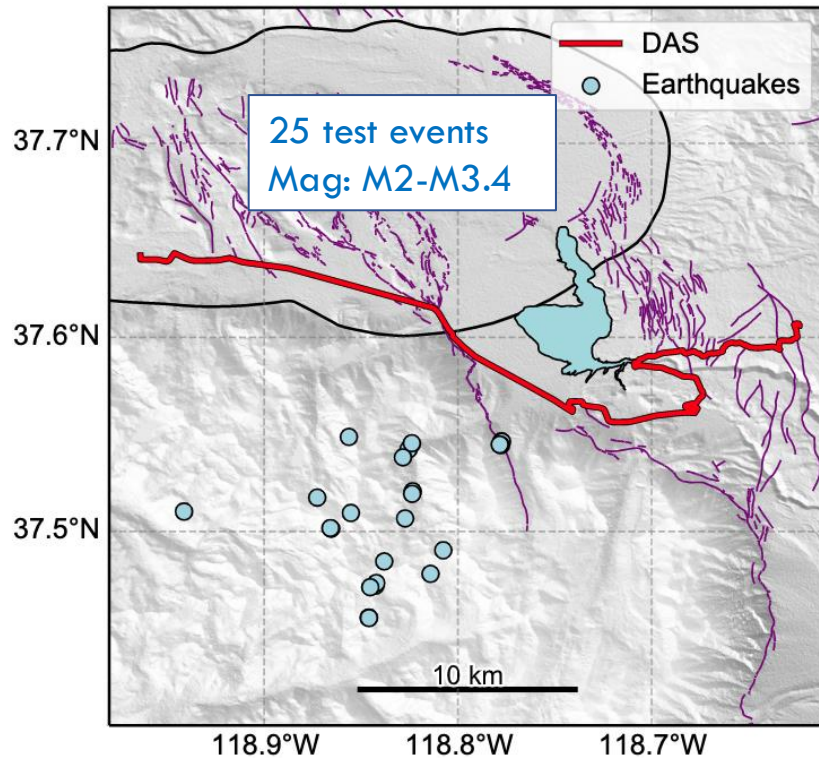
First column vector of $\mathbf{U}$ matrix: $\tilde{\mathbf{u}}_k^{<kk>} = A_k \tilde{\mathbf{p}}_k$ **Ambiguity of -1**

When $k=l+1$: $sign(A_k A_l) = \frac{sign(\tilde{\mathbf{u}}_k^{<kk>} \cdot \tilde{\mathbf{u}}_l^{<ll>})}{sign(\tilde{\mathbf{u}}_k^{<kl>} \cdot \tilde{\mathbf{v}}_l^{<kl>})}$

Summary: workflow for FM Inversion with DAS polarities

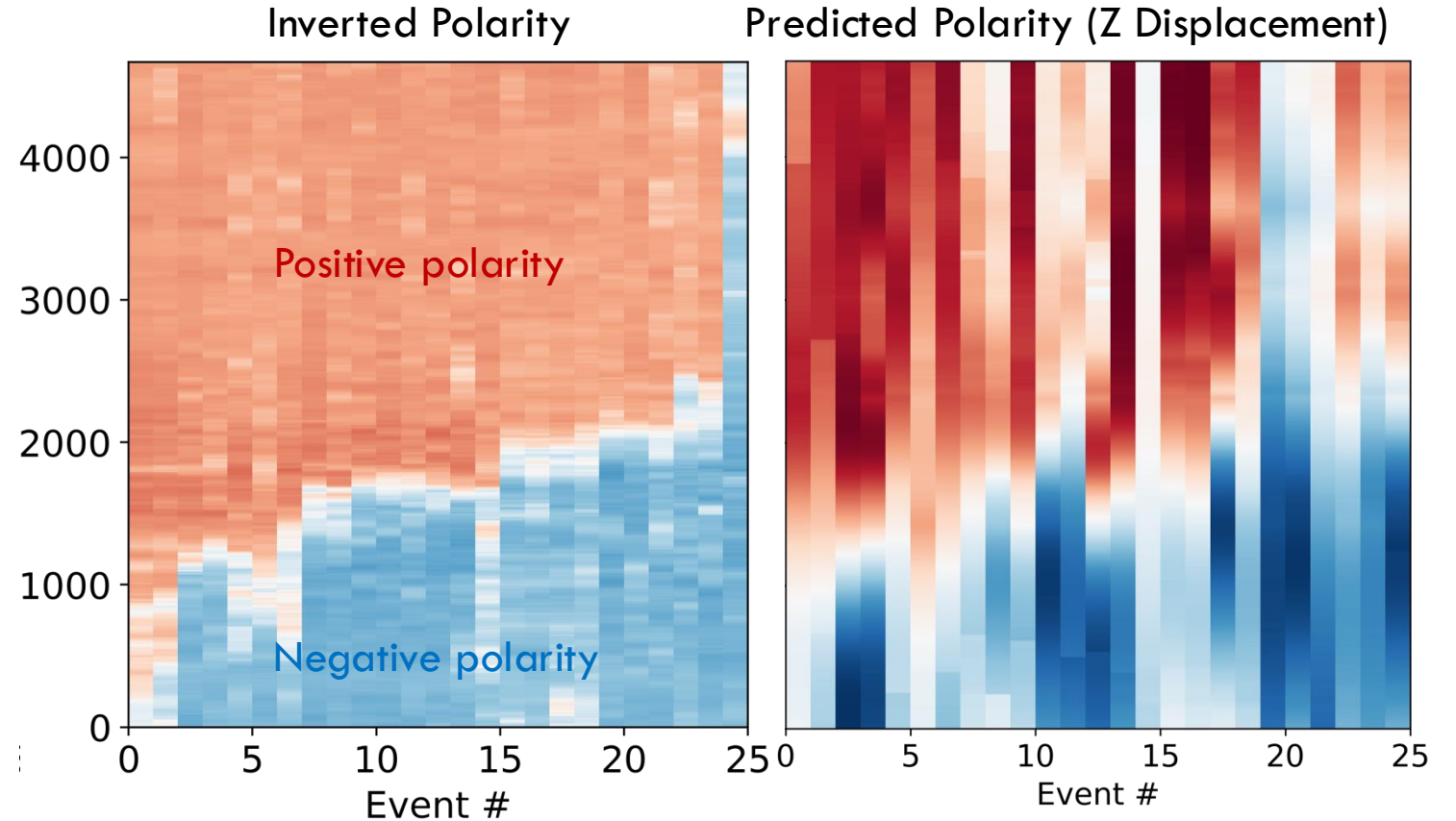


Application: Mammoth south DAS array



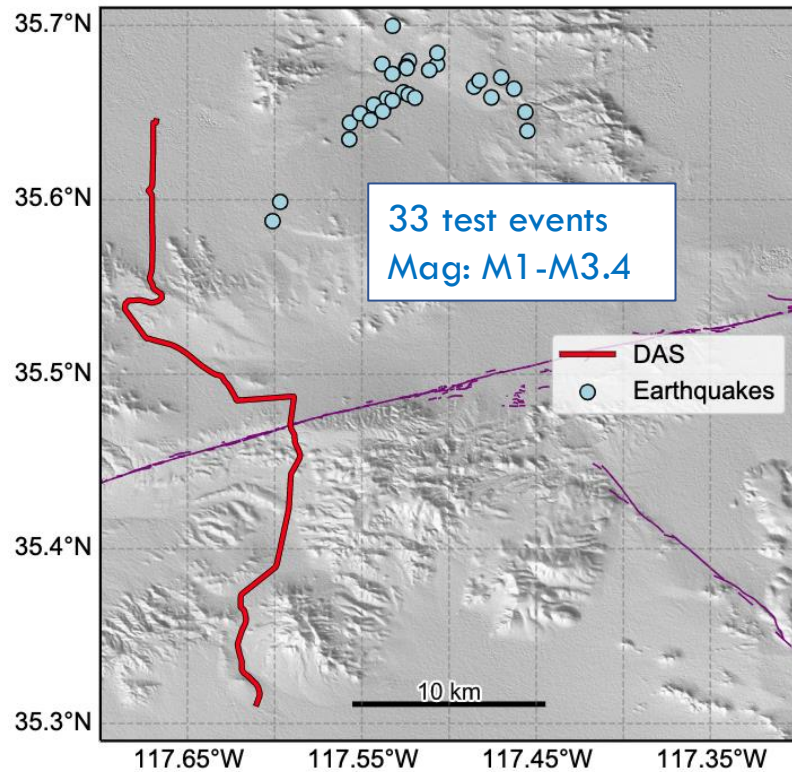
- **Mammoth South DAS array**

- 200 Hz sampling
- 5000 channels
- 10-meter channel spacing

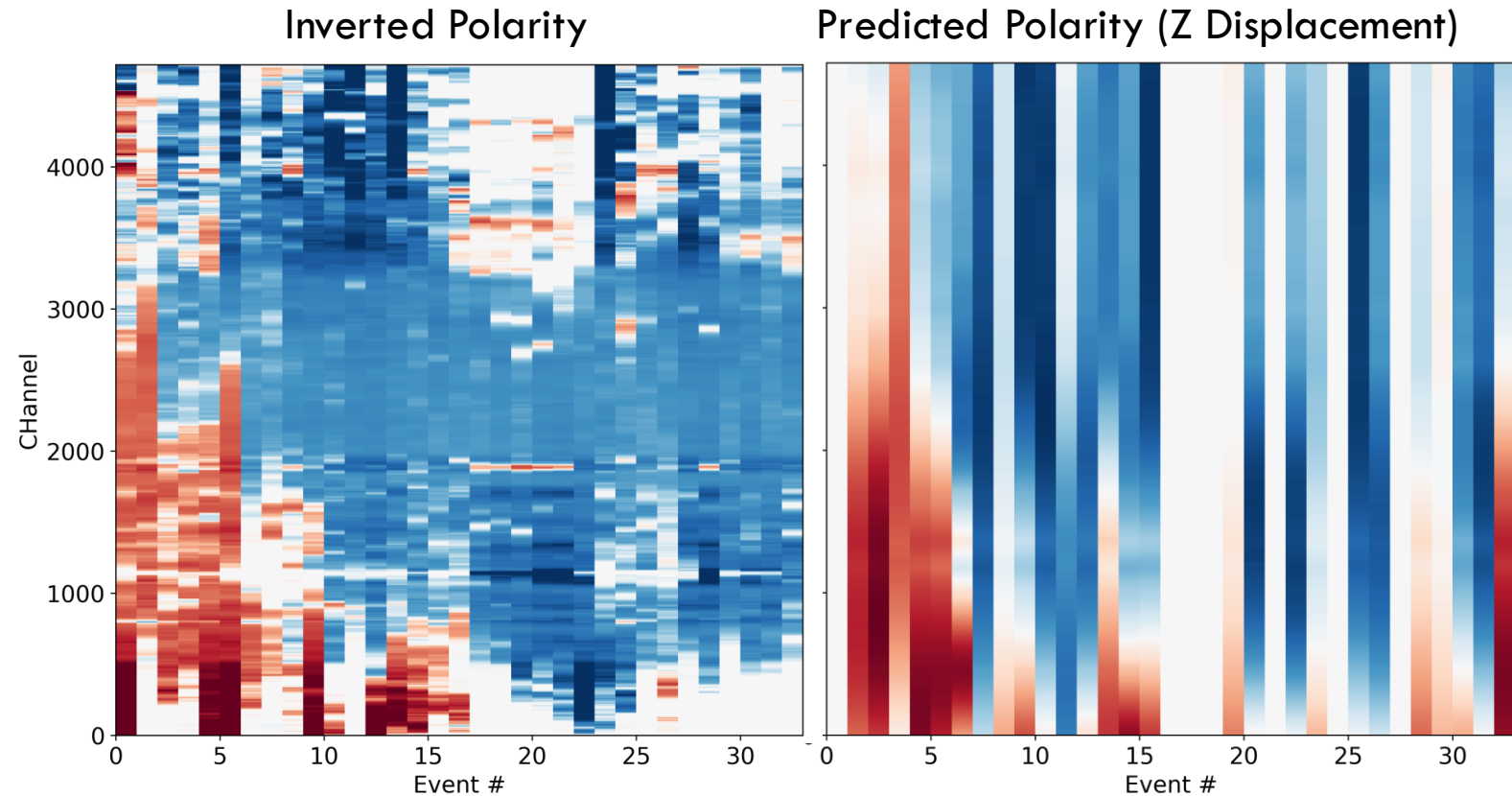


- Extract P phase window...
- Perform CC and MCCC picking...
- ~3 million relative measurements ...

Application: Ridgecrest south DAS array



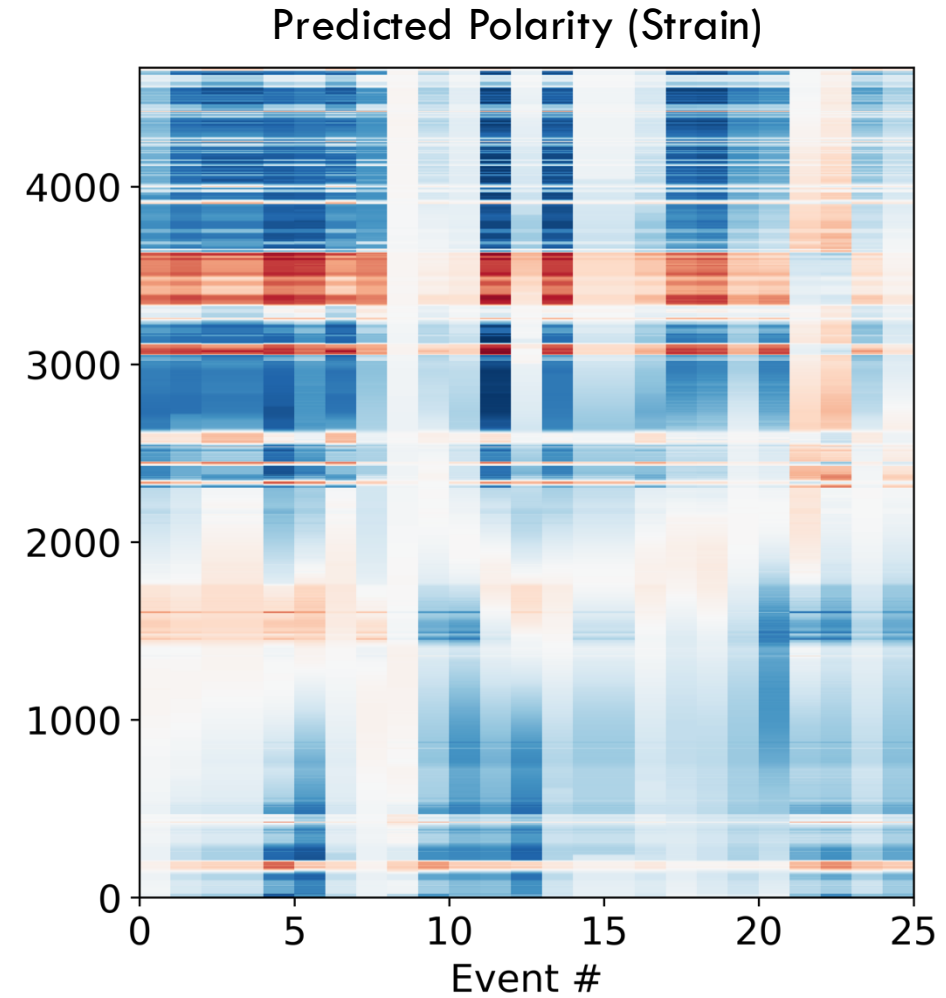
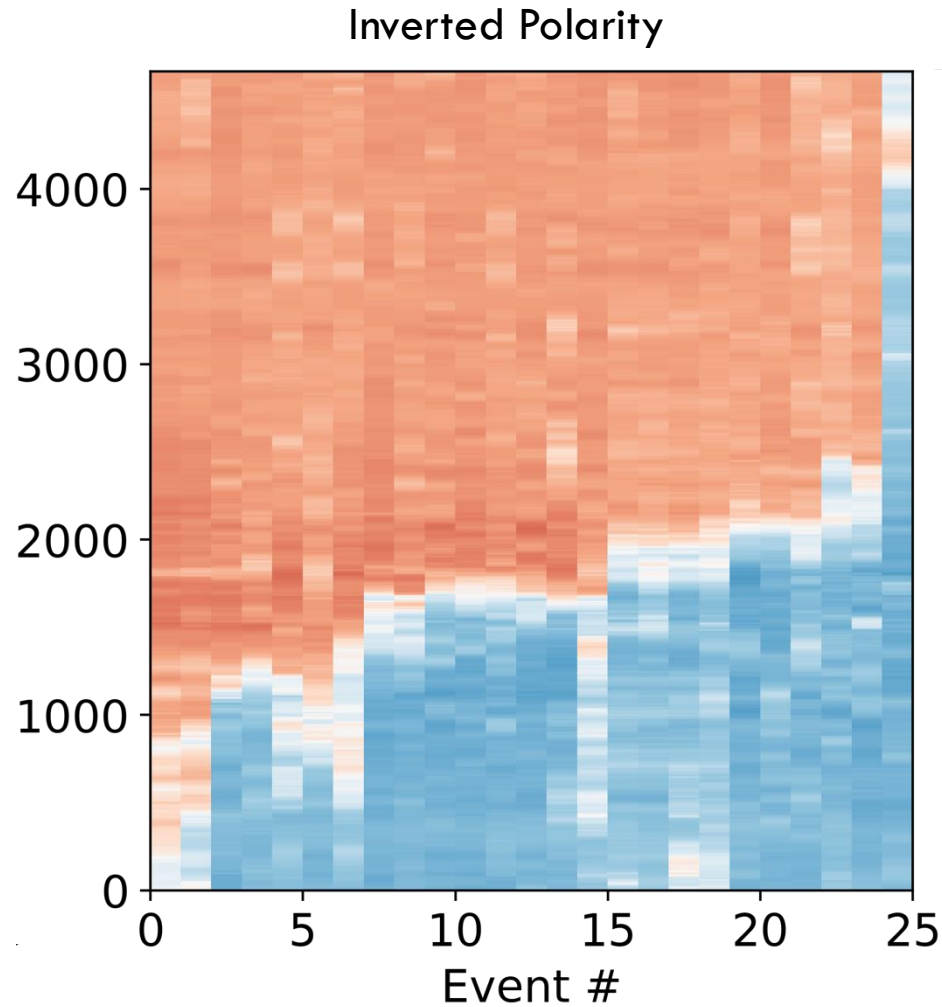
- **Ridgecrest South DAS array**
 - 200 Hz sampling
 - 5000 channels
 - 10-meter channel spacing



But our instrument is recording the strain...

Inverted polarities: **Displacement** or **Strain**?

Long Valley Caldera DAS array

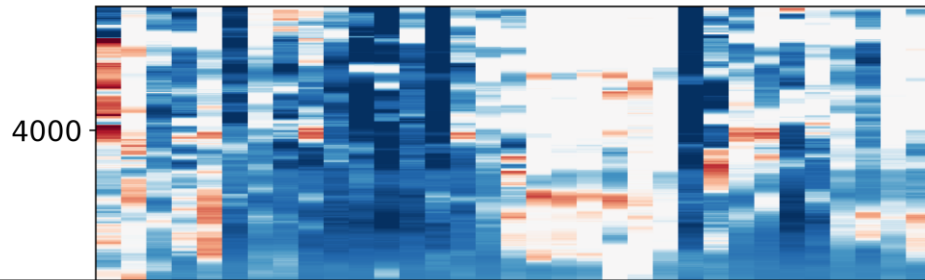


$$\epsilon_{xx} = -\frac{1}{c} \dot{u}_x$$

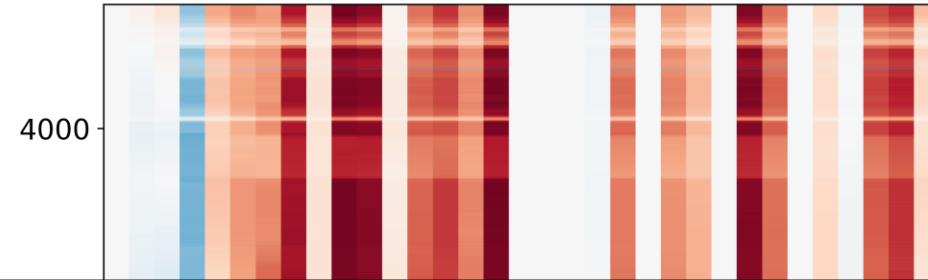
Inverted polarities: **Displacement** or **Strain**?

Ridgecrest South DAS array

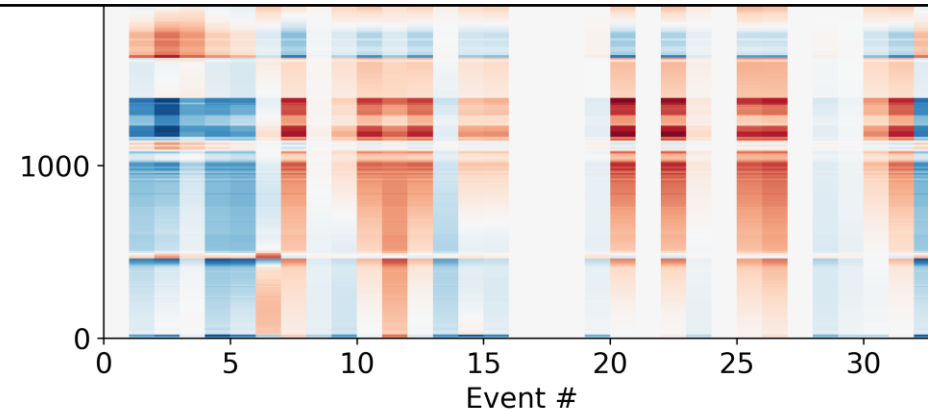
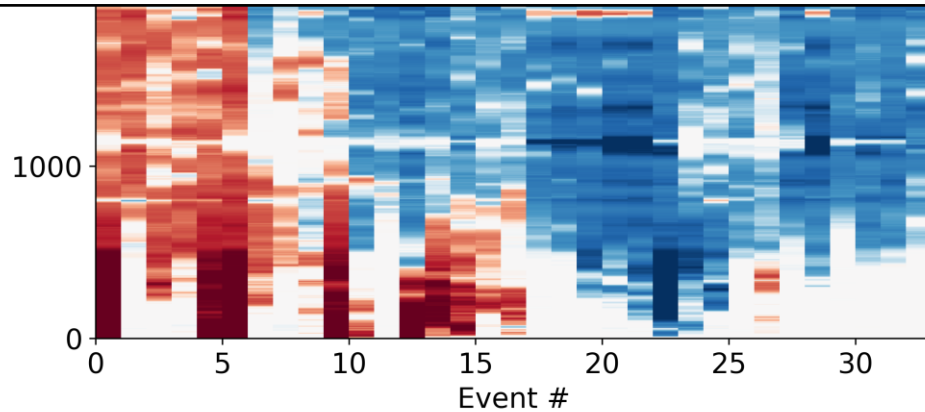
Inverted Polarity



Predicted Polarity (Strain)



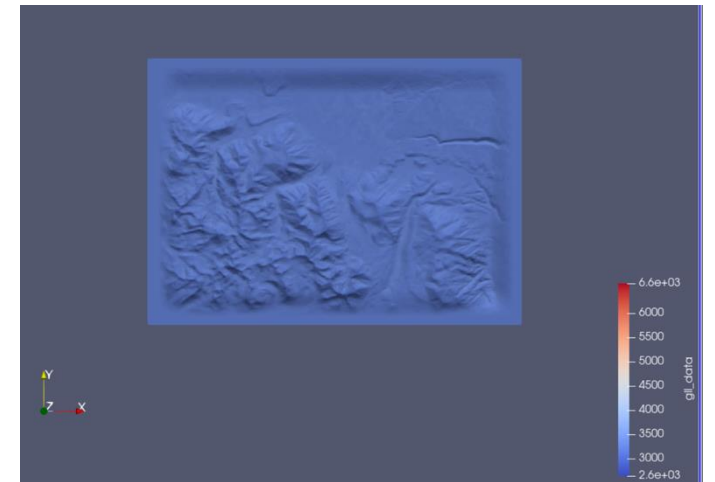
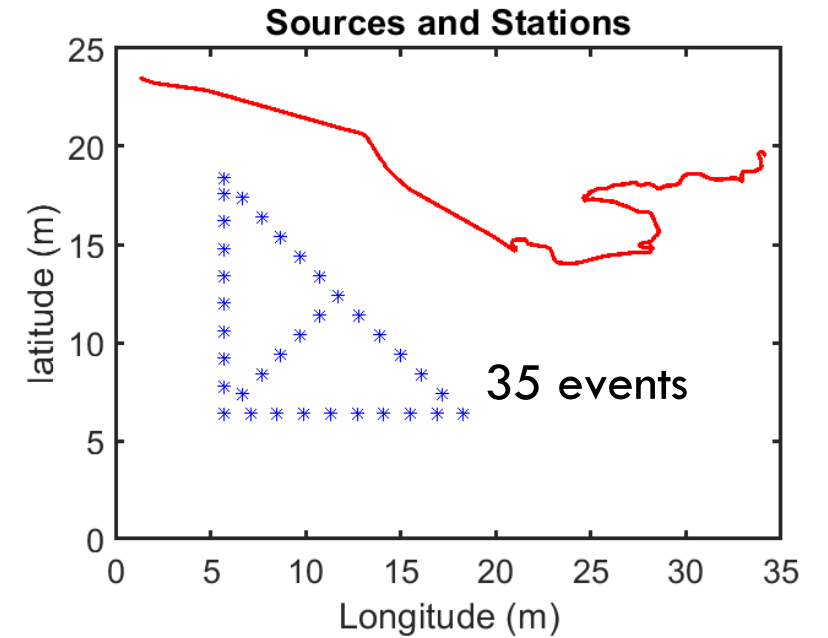
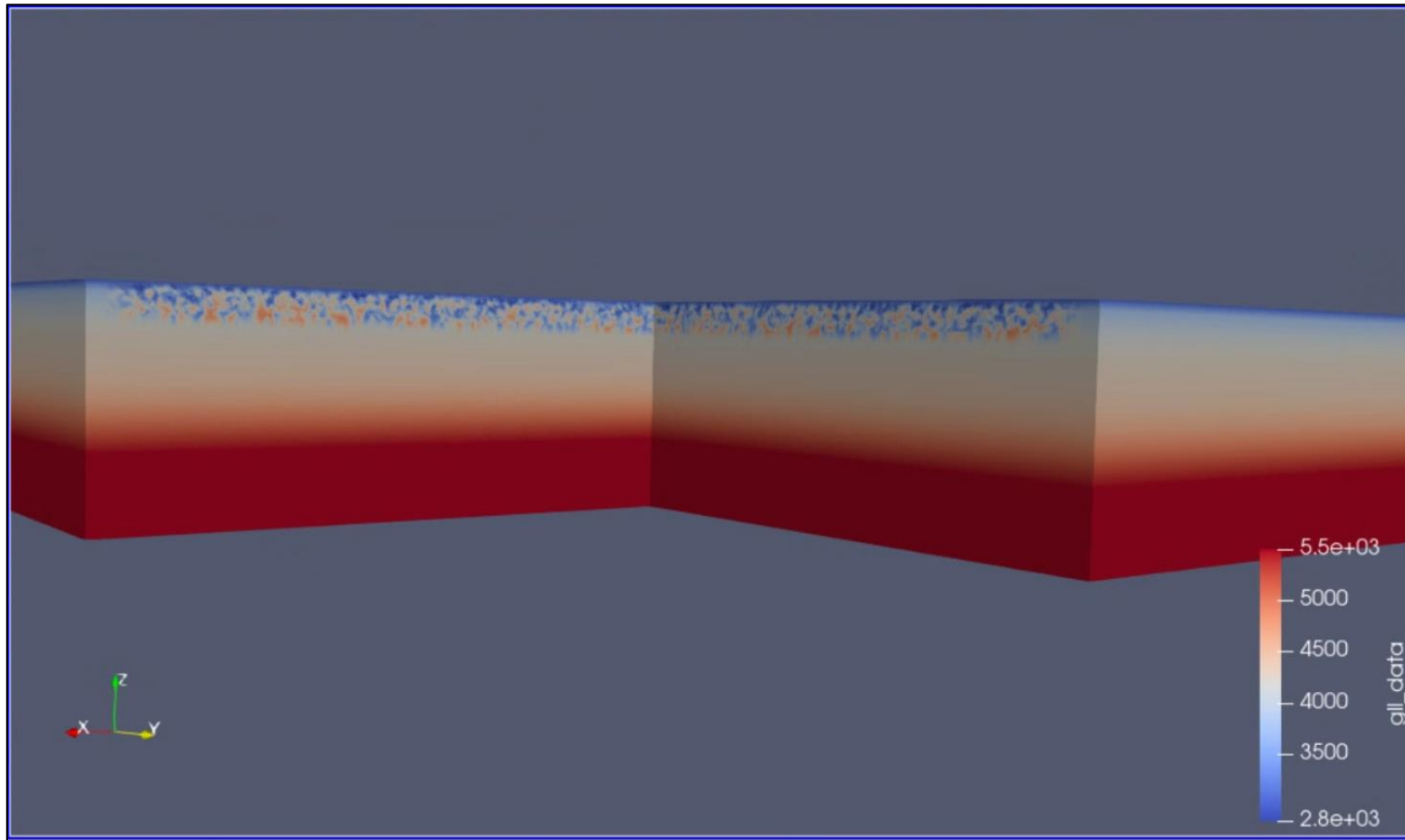
The polarities inverted from the relative measurements directly correspond to vertical displacement polarity!



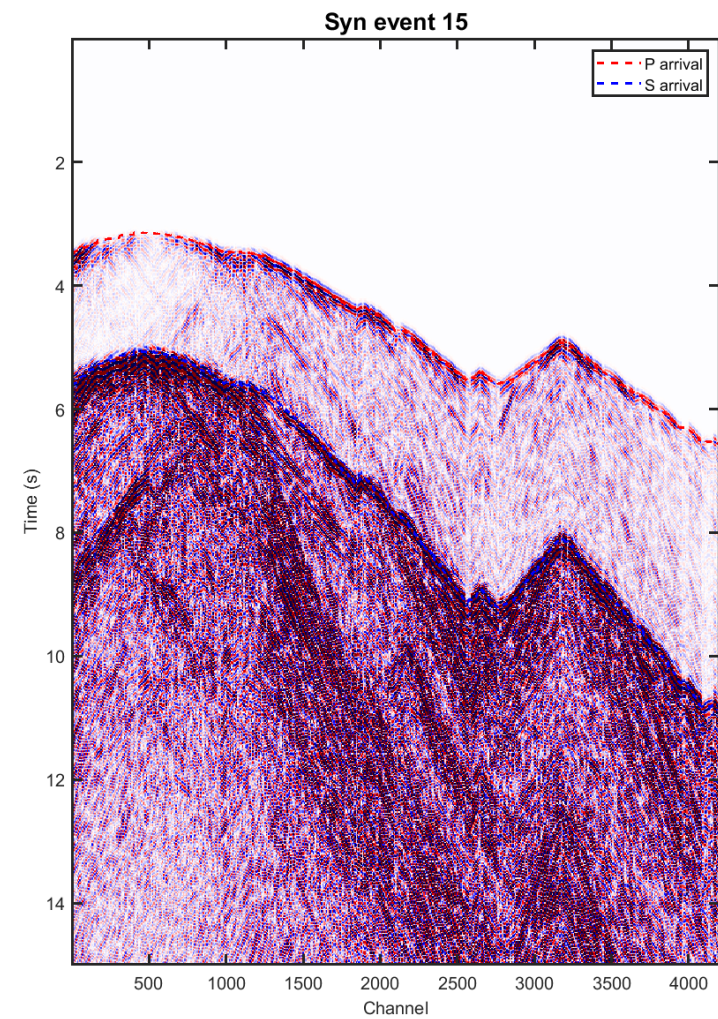
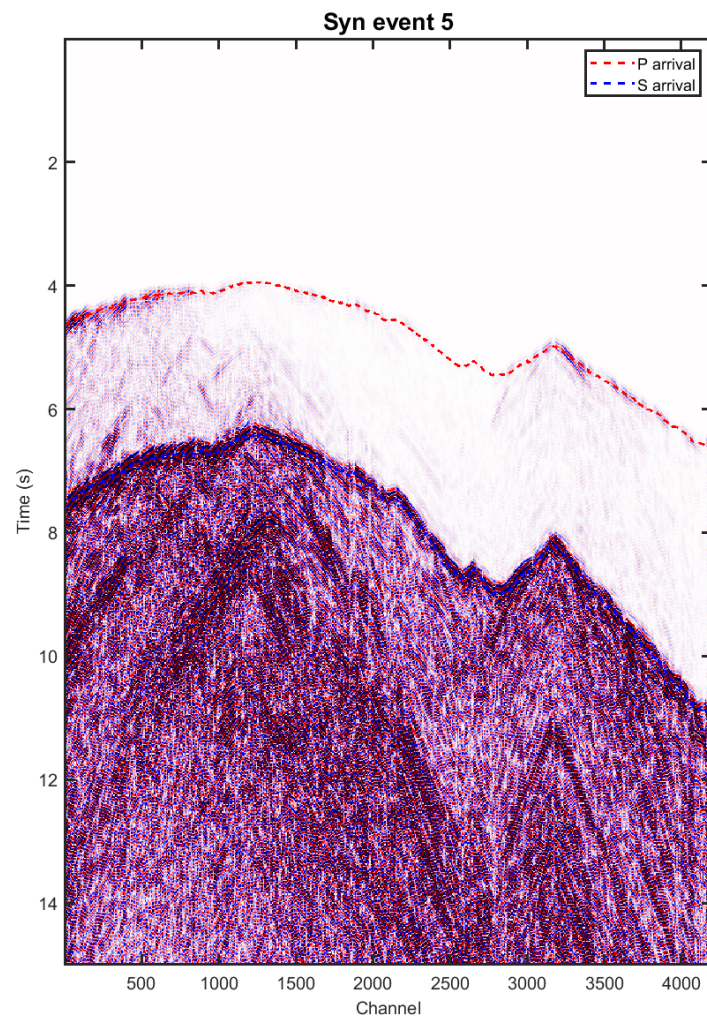
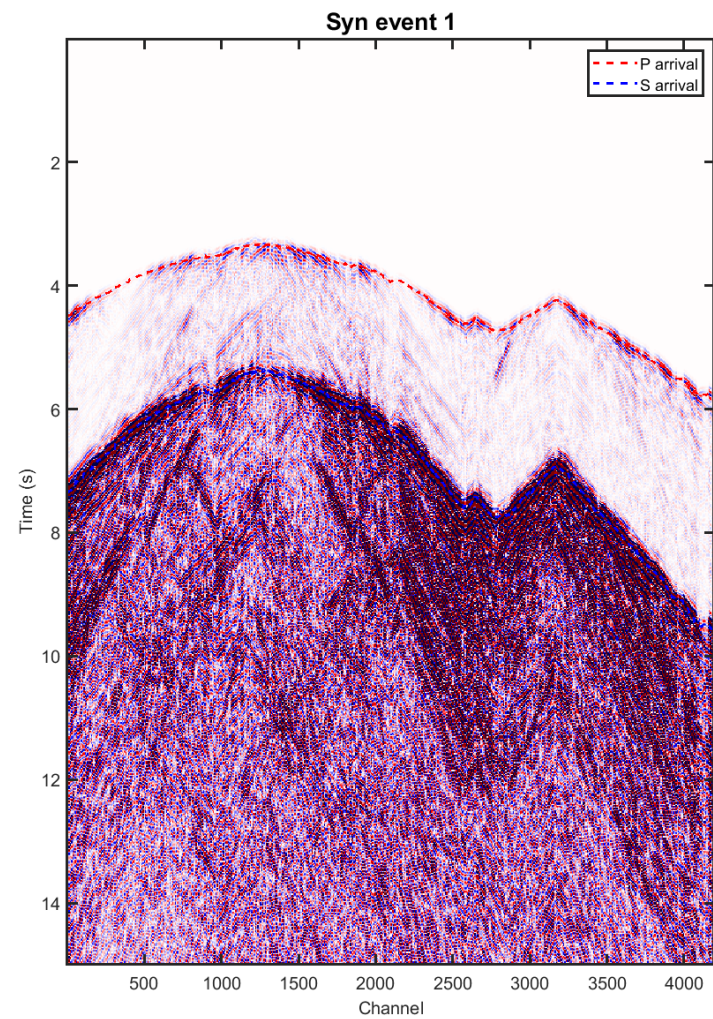
$$\epsilon_{xx} = -\frac{1}{c} \dot{u}_x$$

Reason: near-surface scatterings

Test with Specfem3d simulations: model near-surface scatterings with random shallow velocity anomalies

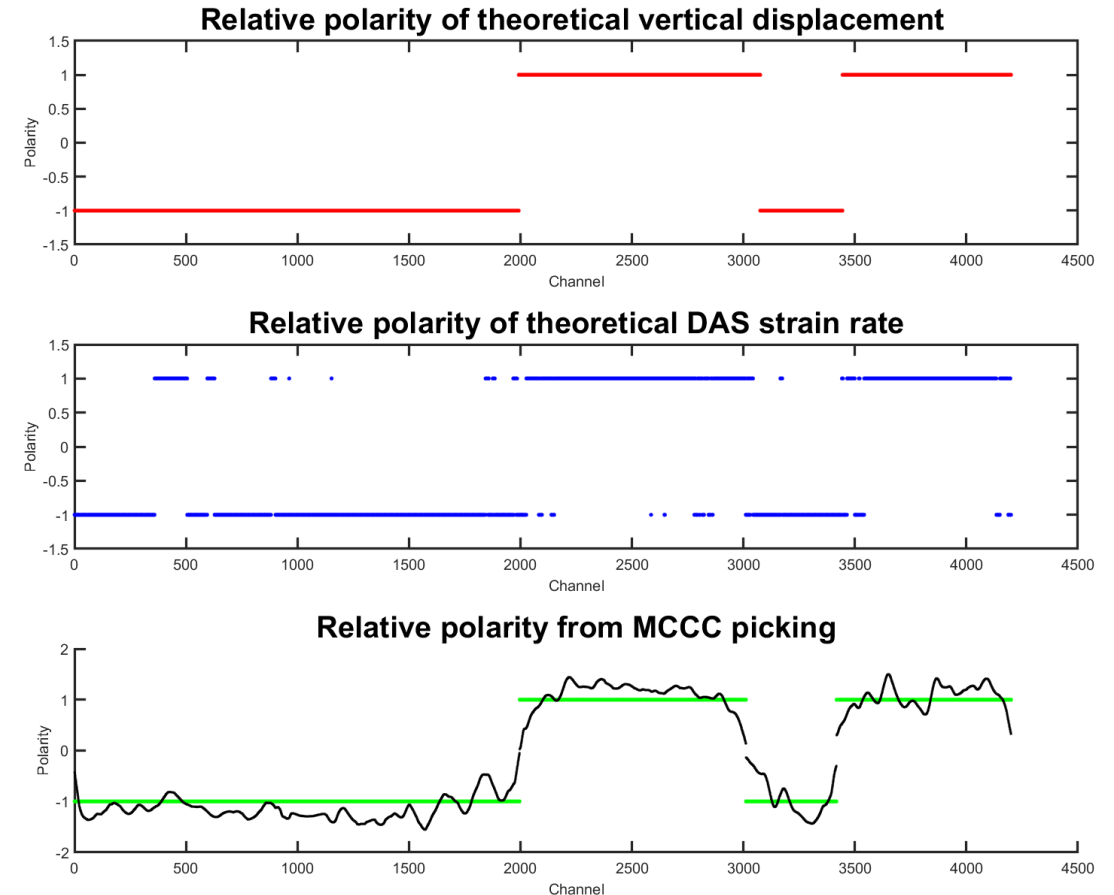
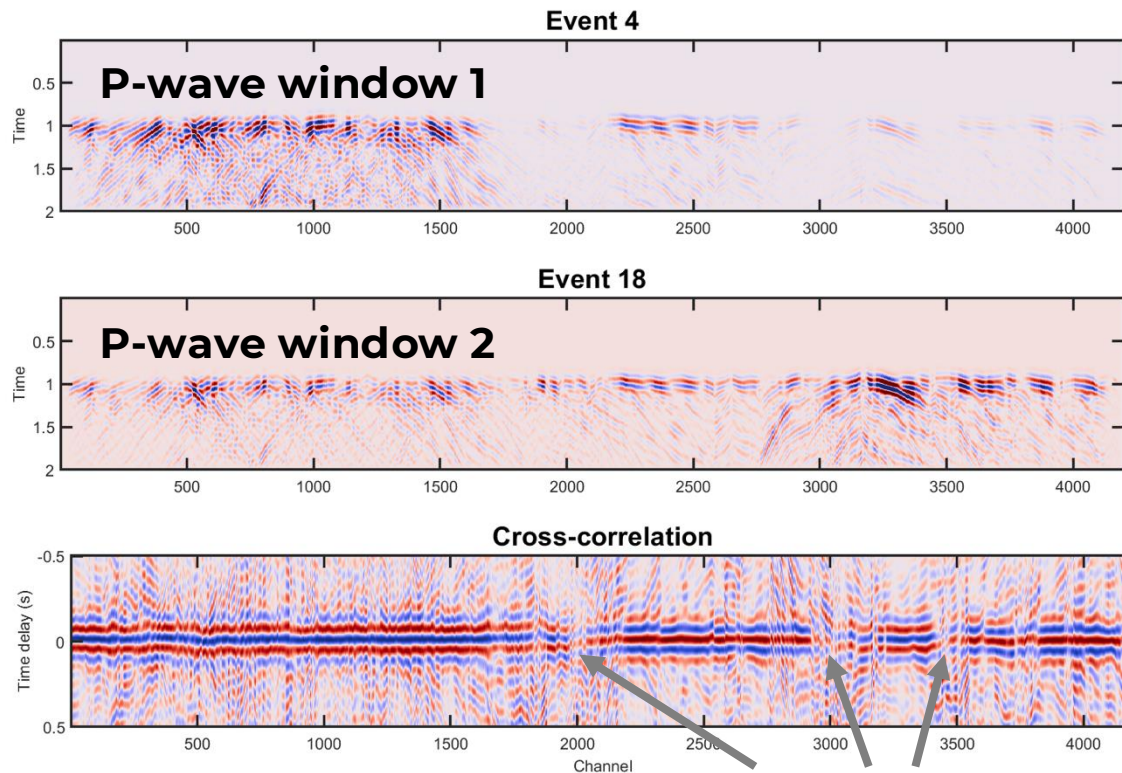


Synthetic DAS strain rate recordings



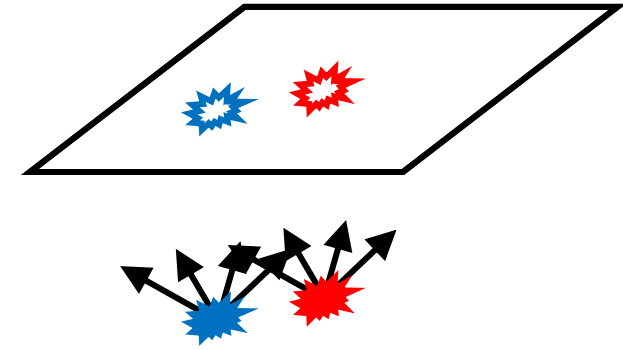
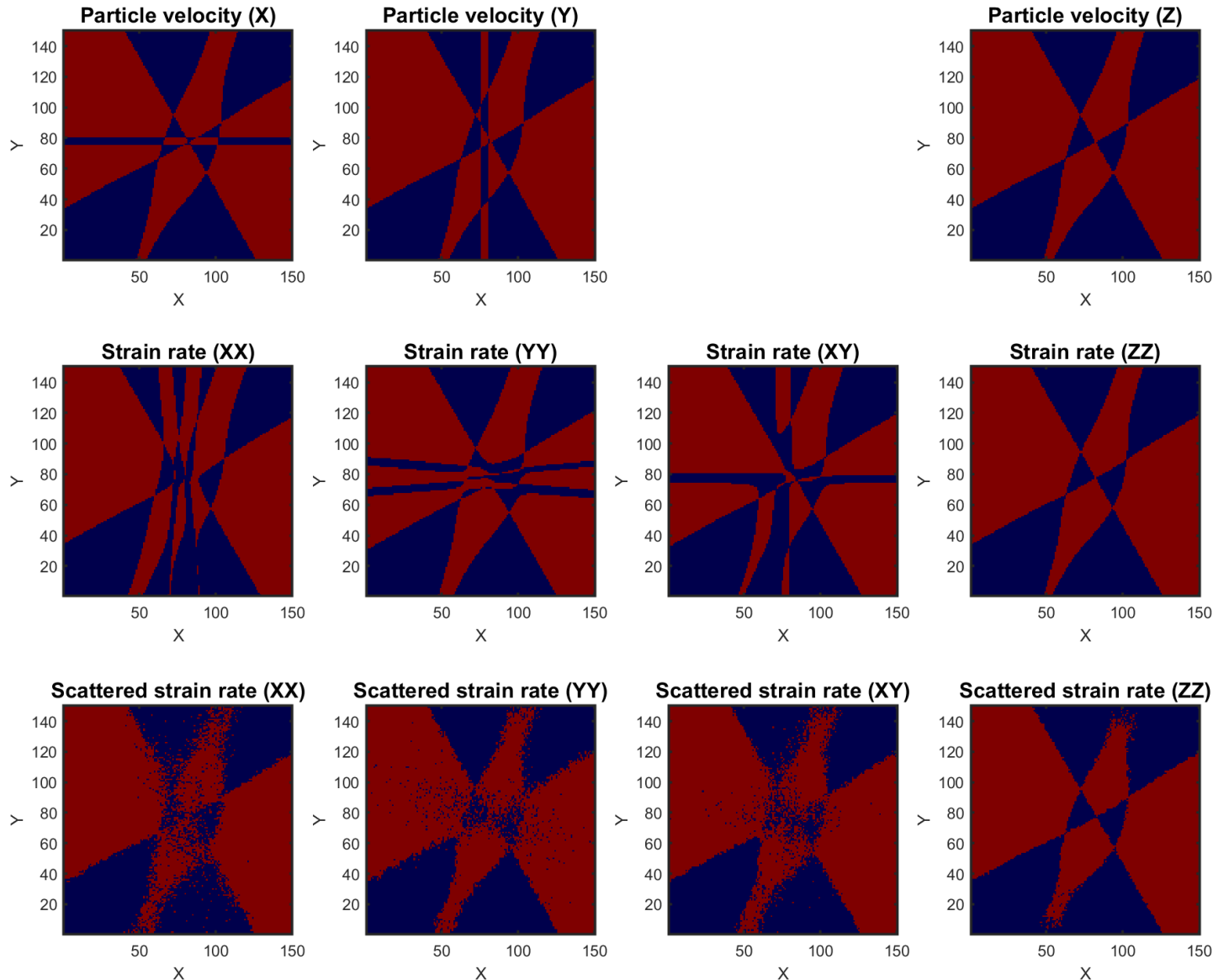
CC between two synthetics P-wave windows

- P-wave window dominated by scatterings.
- Cross-correlation extracts the relative polarity consistent with vertical displacement.
- Scattering preserves the polarity along ray path, which is nearly vertical due to low velocities in shallow surface.



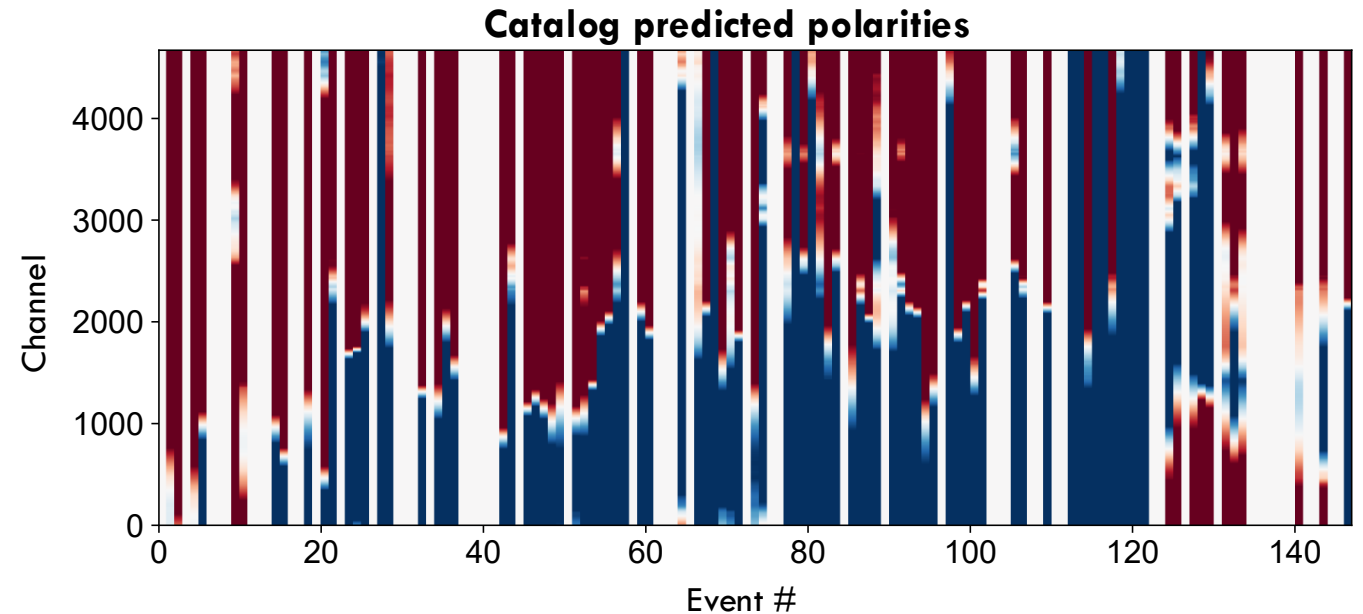
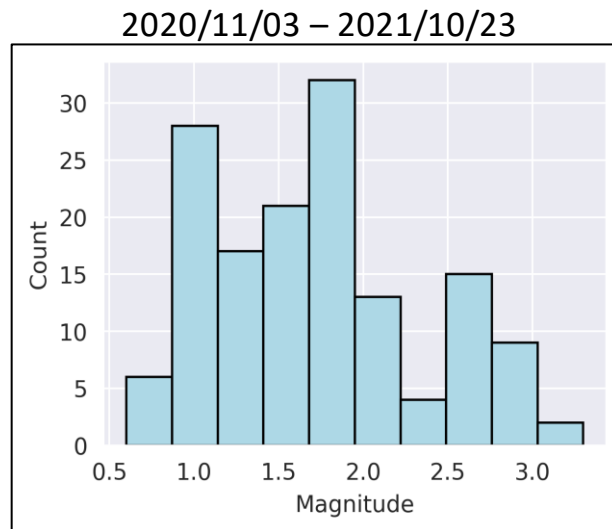
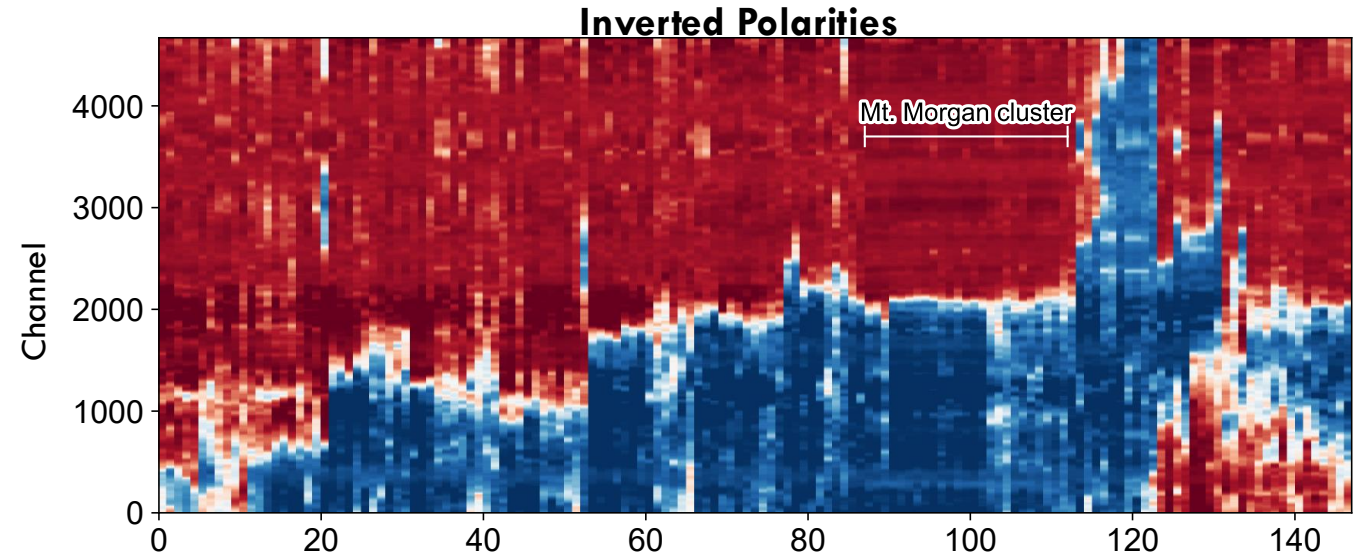
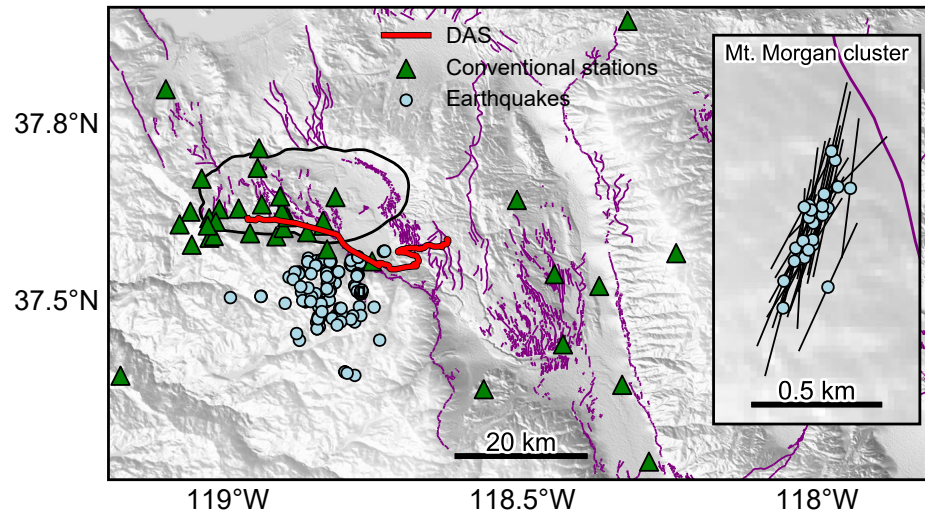
What do $\text{sign}(\text{CC})$ between two events P windows represent?

Relative P-wave Polarity Comparison



- Theoretical particle velocity and strain rate exhibit different relative polarity patterns in the horizontal components.
- However, synthetic results show that cross-correlation extracts consistent relative polarity across different components.
- Near-surface scattering preserves polarity consistent with the Z-component of both particle velocity and strain rate.

Improve earthquake focal mechanisms with DAS polarities



(Li et al., 2023)

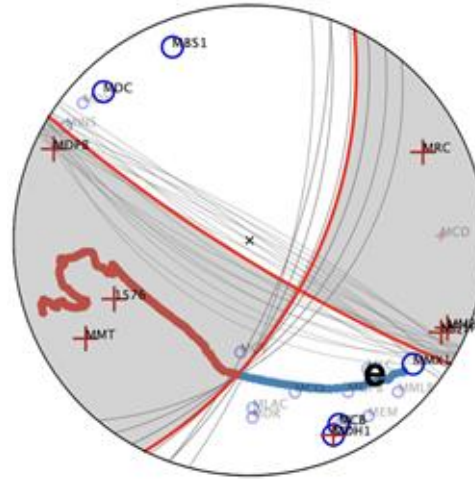
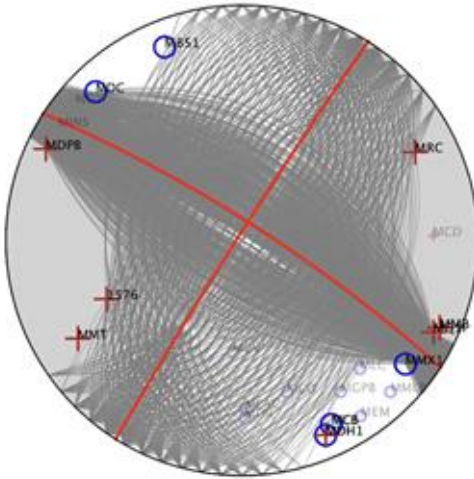
White gap means no available focal mechanism from the NCEDC catalog

Improve earthquake focal mechanisms with DAS polarities

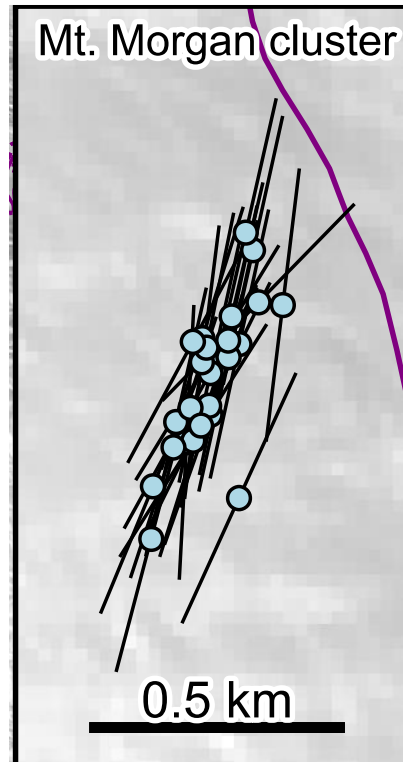
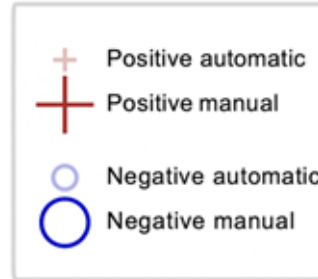
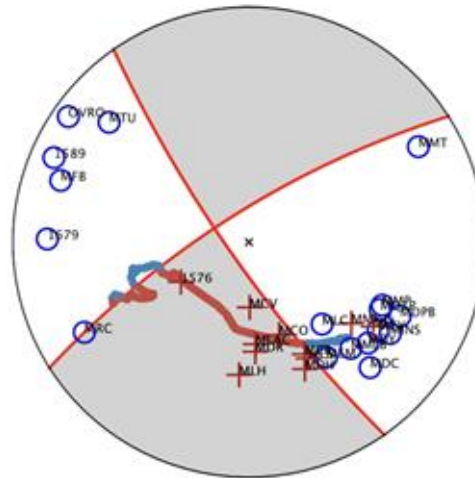
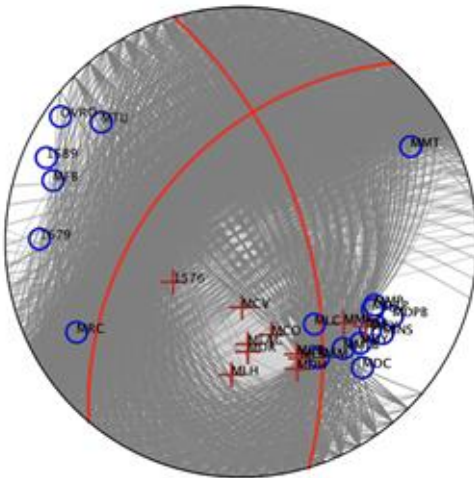
Conventional sensor only

Conventional sensor + DAS

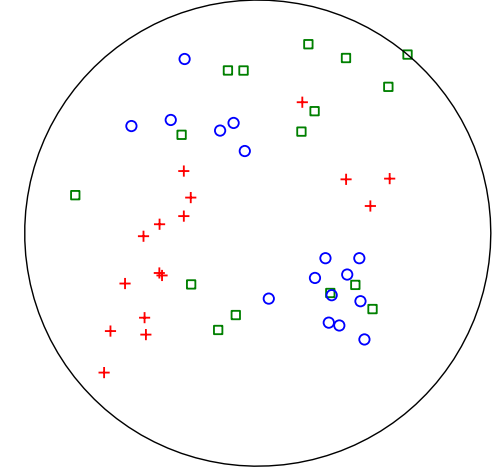
EQ1



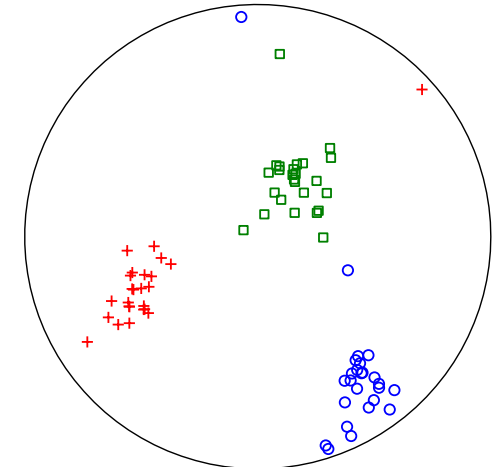
EQ2



Catalog

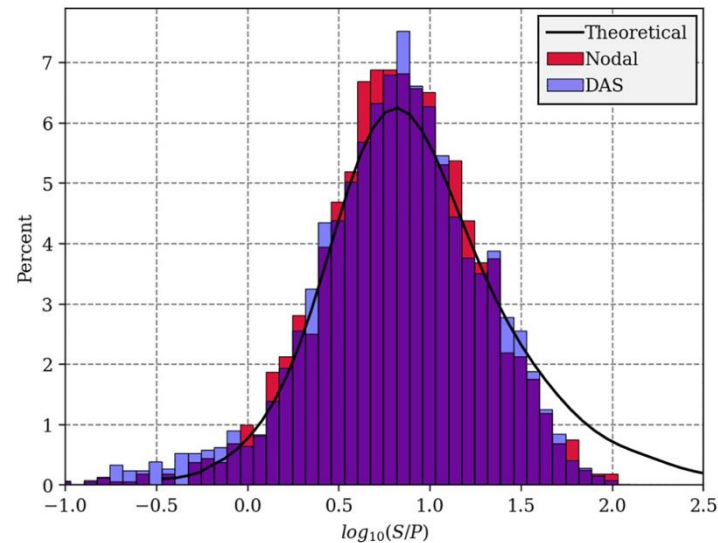
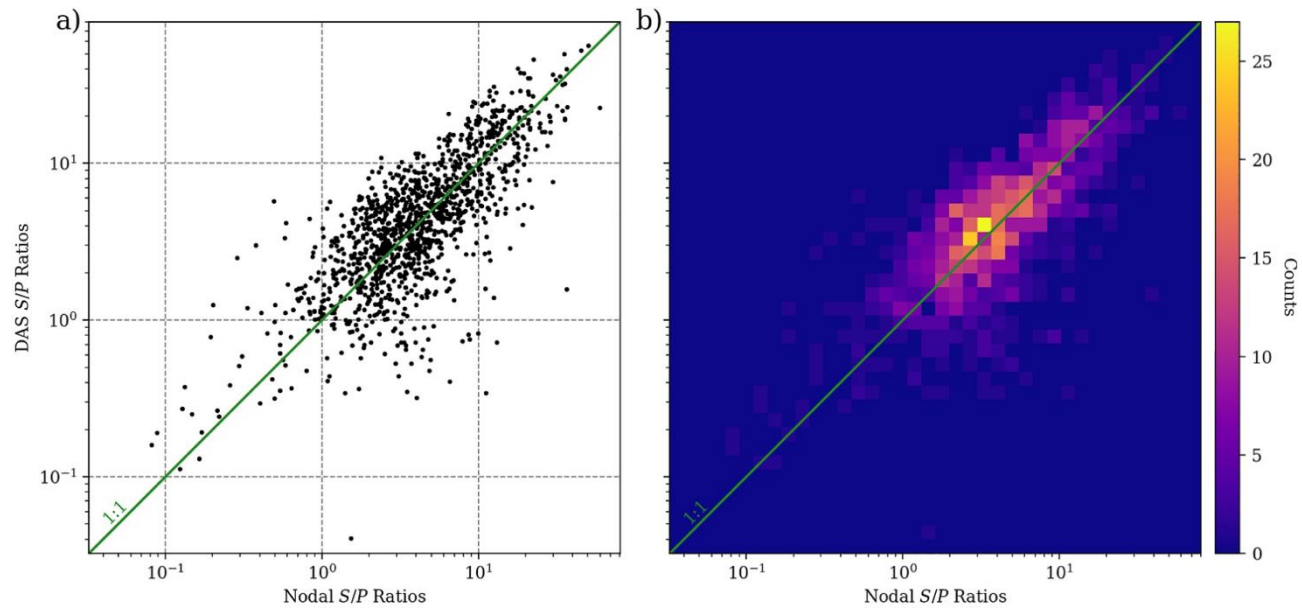


Joint



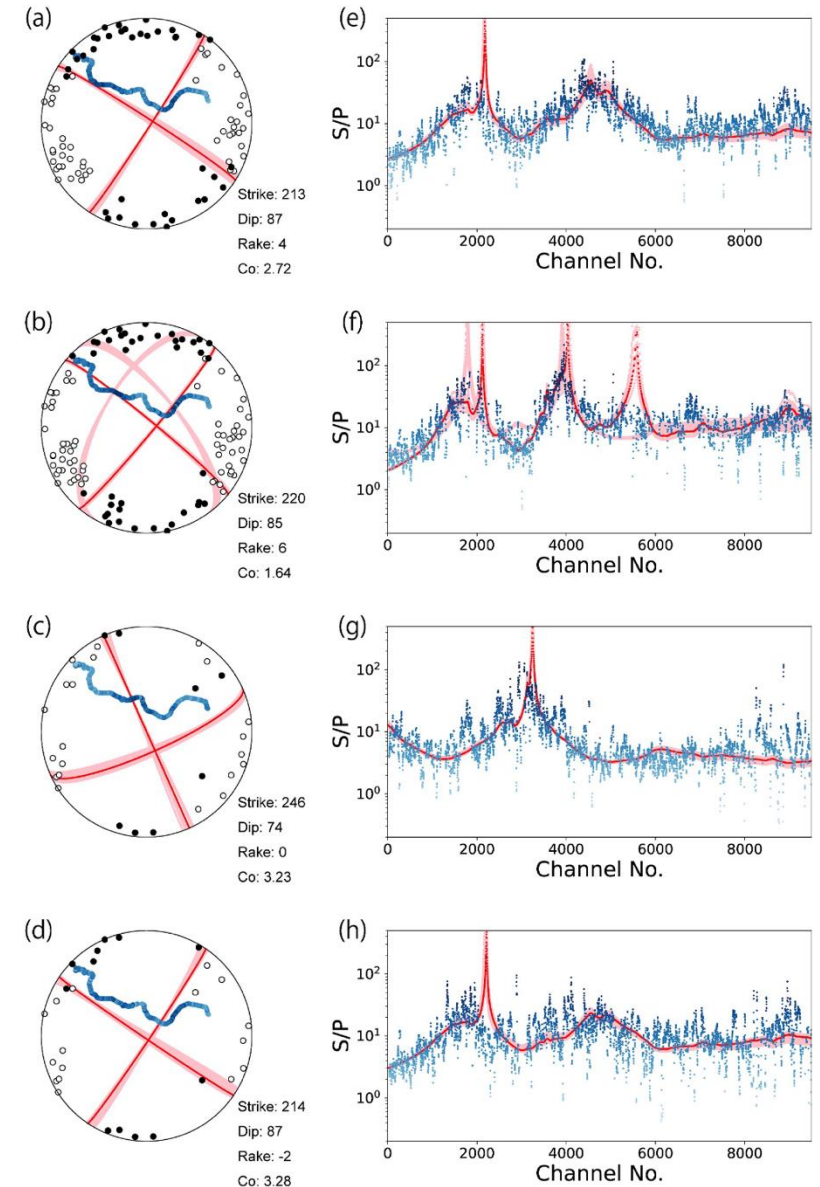
(Li et al., 2023)

What about amplitude information?



Again, DAS is recording the longitudinal strain...

(Skoumal et al., 2025)



(Funabiki and Miyazawa 2025)

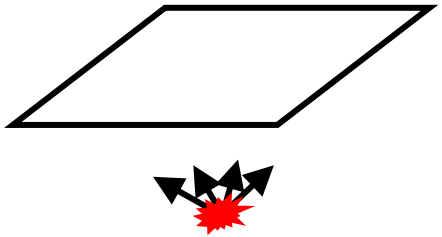
Theoretical S/P ratio for velocity and strain rate

Far-field SP ratio of:

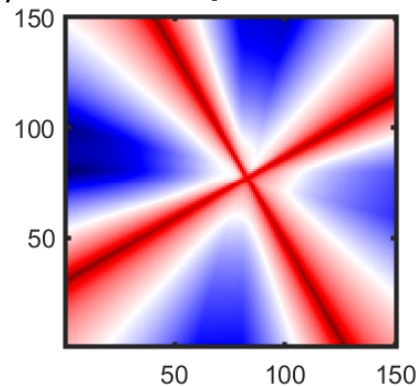
- Particle velocity (vector L2 norm)
- Strain rate (tensor L2 norm)
- S/P ratio differs by a constant factor: $V_P/(V_S\sqrt{2})$.

$$\frac{\|v_i^S(\mathbf{r}, t)\|}{\|v_i^P(\mathbf{r}, t)\|} = \frac{V_P^3}{V_S^3} \sqrt{\frac{\gamma_k \gamma_l M_{pk} M_{pl} - (\gamma_k \gamma_l M_{kl})^2}{(\gamma_k \gamma_l M_{kl})^2}}$$

$$\frac{\|\dot{\epsilon}_{i,j}^S(\mathbf{r}, t)\|}{\|\dot{\epsilon}_{i,j}^P(\mathbf{r}, t)\|} = \frac{V_P^4}{V_S^4} \frac{1}{\sqrt{2}} \sqrt{\frac{\gamma_k \gamma_l M_{pk} M_{pl} - (\gamma_k \gamma_l M_{kl})^2}{(\gamma_k \gamma_l M_{kl})^2}}$$

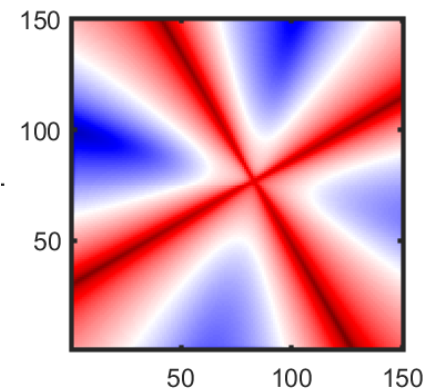


S/P velocity norm ratio



$$\frac{V_P}{\sqrt{2}V_S} \cdot \frac{\|\mathbf{v}^S\|}{\|\mathbf{v}^P\|} = \frac{\|\dot{\epsilon}^S\|}{\|\dot{\epsilon}^P\|}$$

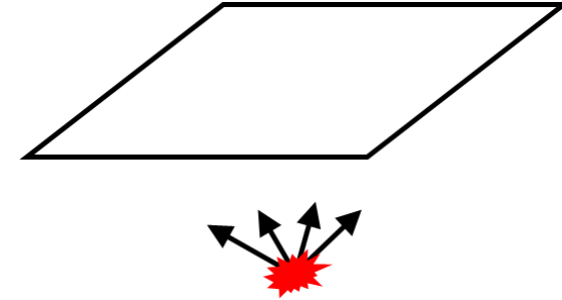
S/P strain rate norm ratio



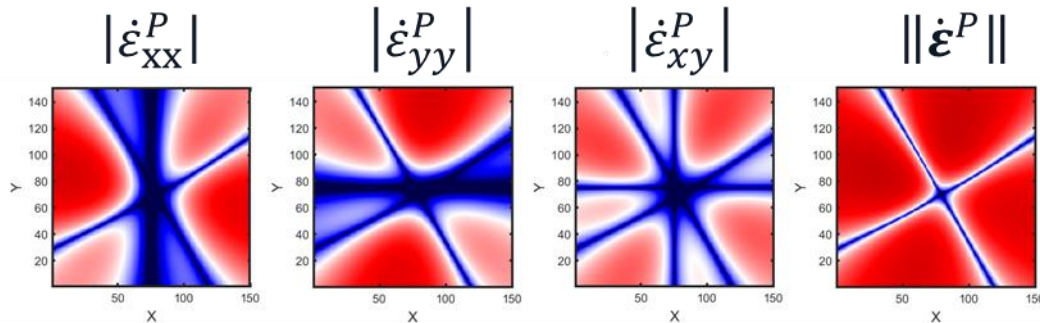
Effect of scatterings on DAS strain-rate amplitudes

- DAS measures longitudinal strain along the fiber.
- For P waves, scatterings make the horizontal components match the pattern of theoretical tensor norm amplitude.
- For S waves, scatterings have similar but weaker effects.

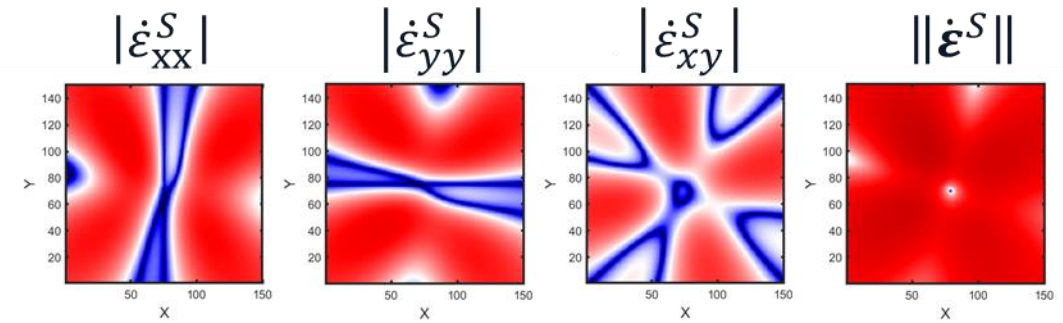
2D-surface recording



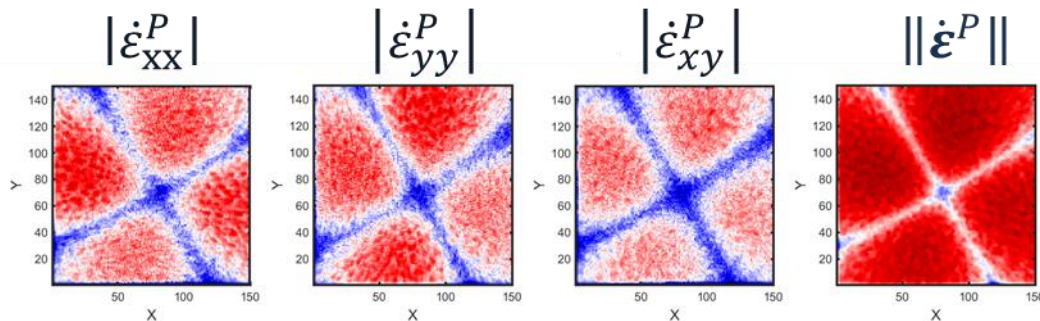
Theoretical **P amplitude** patterns



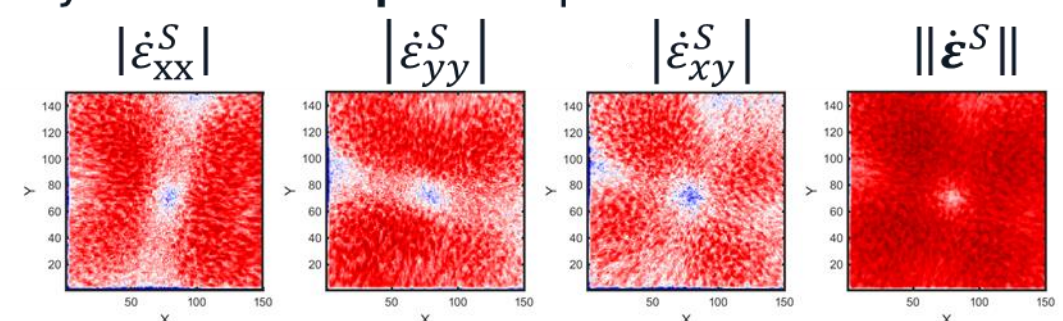
Theoretical **S amplitude** patterns



Synthetic **P amplitude** patterns with **scattering**



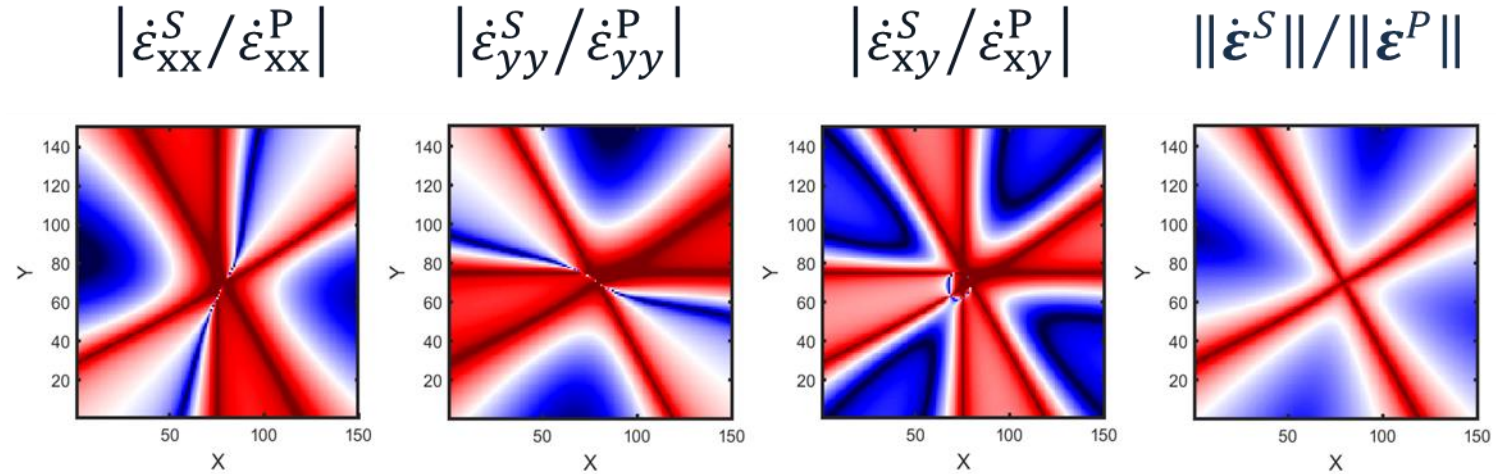
Synthetic **S amplitude** patterns with **scattering**



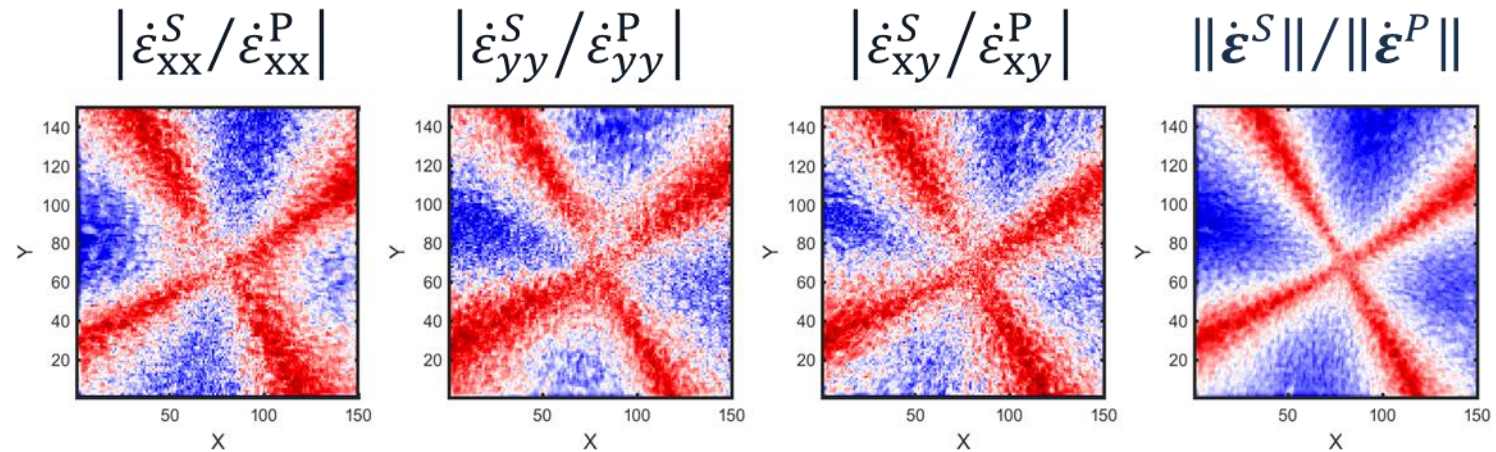
Effect of scatterings on DAS strain-rate S-P ratios

- The theoretical S-P ratios between single components show significant difference.
- Scatterings makes the synthetic S-P ratios similar to the theoretical tensor-norm S-P ratio, which shares the same pattern as the traditional displacement S-P ratios.

Theoretical S-P ratios patterns



Synthetic S-P ratios patterns with scattering



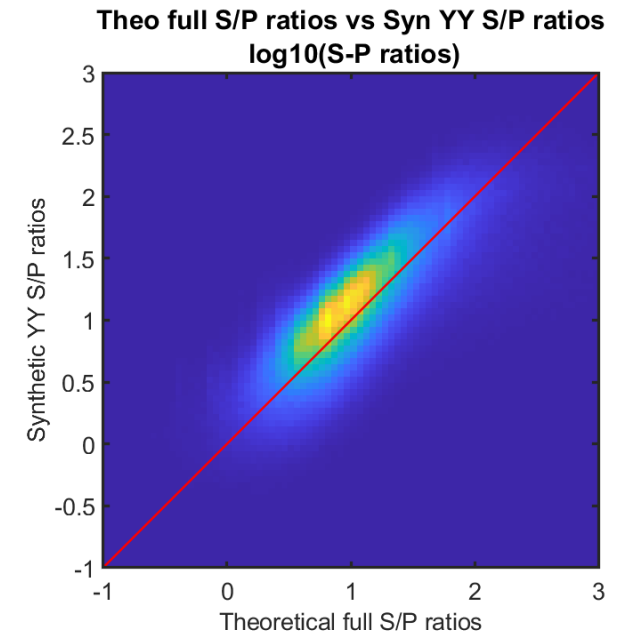
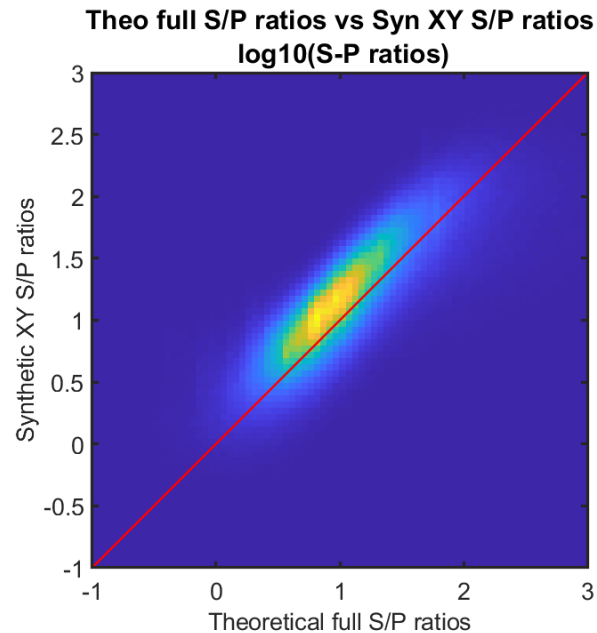
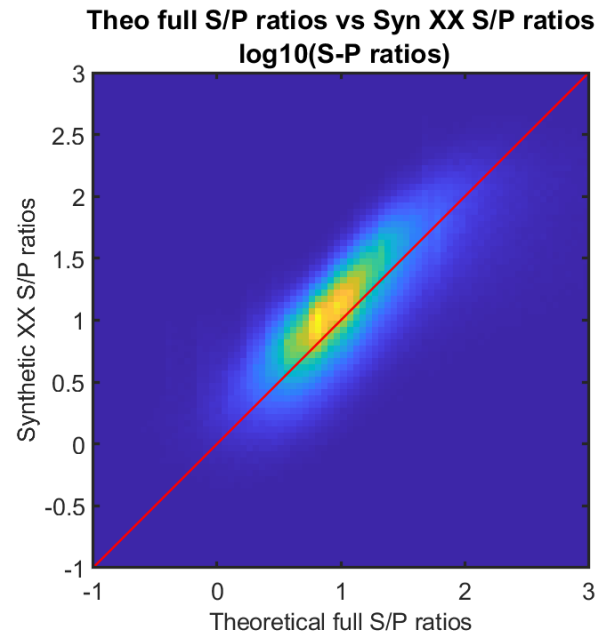
$$\frac{\|\dot{\epsilon}^S\|}{\|\dot{\epsilon}^P\|} = \frac{V_P}{\sqrt{2}V_S} \cdot \frac{\|v^S\|}{\|v^P\|}$$

DAS S/P ratios: scattering effect

SPECFEM3D synthetic with: near-surface scatterings

Comparison between **synthetic** $\dot{\epsilon}_{xx}$, $\dot{\epsilon}_{xy}$, $\dot{\epsilon}_{yy}$ S/P ratios with the **theoretical** $\|\dot{\epsilon}\|$ S/P ratios;

- Approximately linear relationship (with some deviation)
- The distribution is shifted upward from the diagonal;
- Meaning: single-component (DAS) S/P ratios are systematically higher than the theoretical S/P ratios.

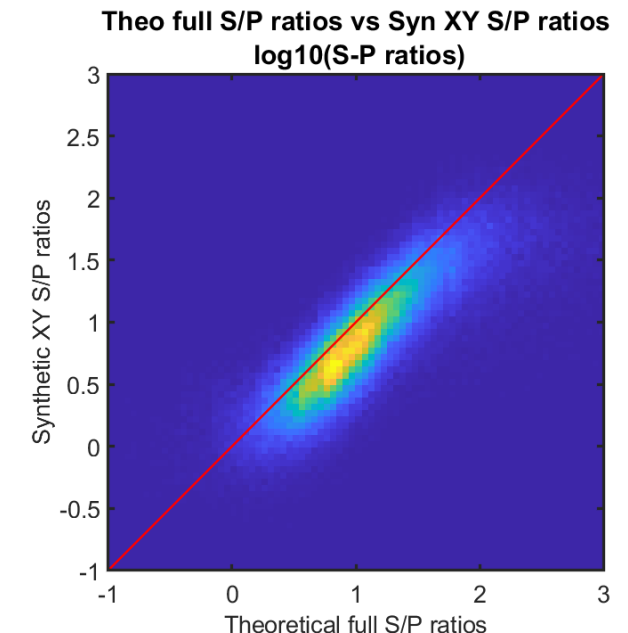
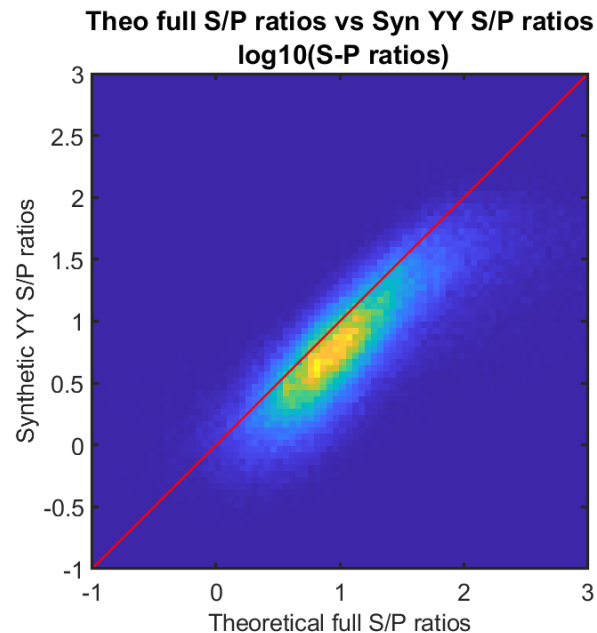
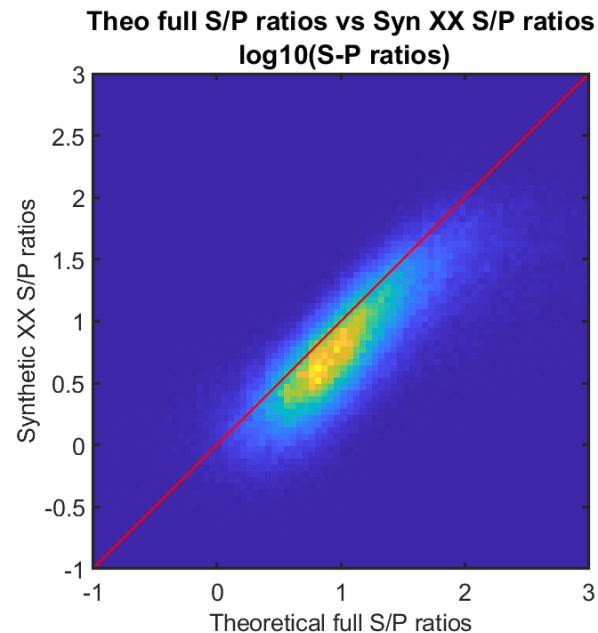
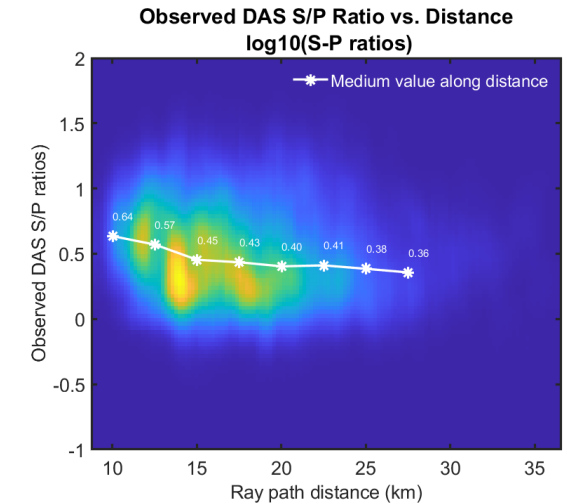


DAS S/P ratios: attenuation effect

SPECFEM3D synthetic with: near-surface scatterings + attenuation

Comparison between **synthetic** $\dot{\epsilon}_{xx}$, $\dot{\epsilon}_{xy}$, $\dot{\epsilon}_{yy}$ S/P ratios with the **theoretical** $\|\dot{\epsilon}\|$ S/P ratios;

- The linear relationship is shifted downwards.
- Q effect distance-dependent (S attenuates more than to P) and requires more care on corrections.



Using DAS S/P amplitude ratios for FM inversion

- S/P ratios on DAS **strain-rate** recordings have similar patterns to S/P ratios on conventional **velocity** recordings due to **near-surface scatterings**.
- However, S/P ratios on DAS is an approximate and shifted version of S/P ratios on velocity recordings due to:
 - The constant shift $V_P/(V_S\sqrt{2})$
 - Near-surface scattering effect
 - Attenuation (which also affects conventional stations)
- The simplest strategy would be applying a constant shift to synthetic S/P ratios to match observed DAS S/P ratios.
 - Synthetic S/P ratios could be either strain-rate or velocity

DASFM: joint focal mechanism inversion

What is DASFM?

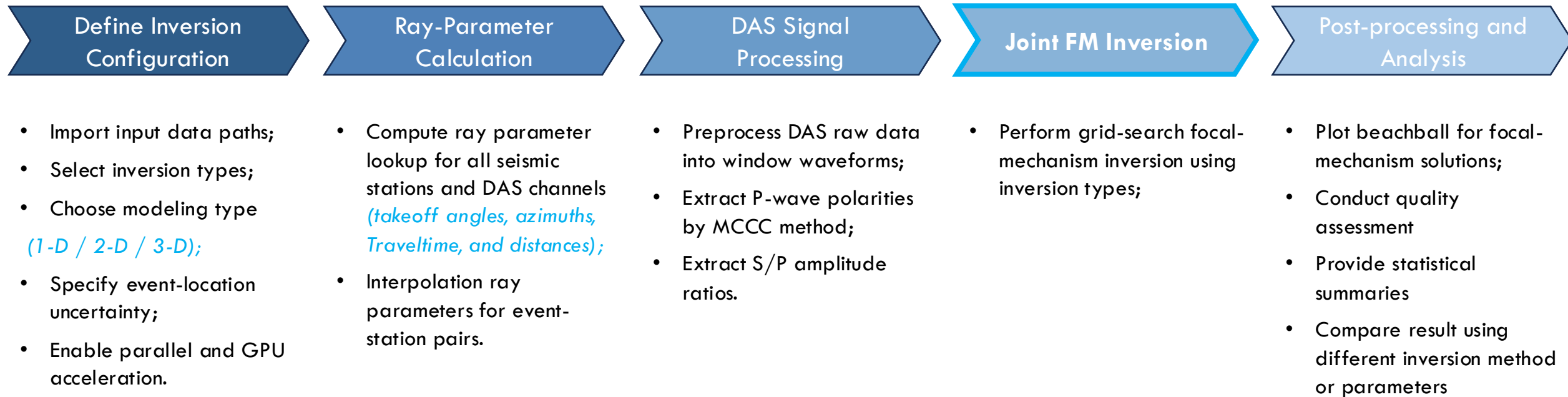
- A framework for earthquake focal mechanism inversion.
- Provides a complete workflow to extract P-wave polarity and S/P ratios from DAS data.
- Performs joint inversion using DAS and conventional seismic station data.

Station polarity	Station S/P	DAS polarity	DAS S/P	Supported
✓				✓
✓	✓			✓
		✓		✓
		✓	✓	✓
✓		✓		✓
✓			✓	✓
✓		✓	✓	✓

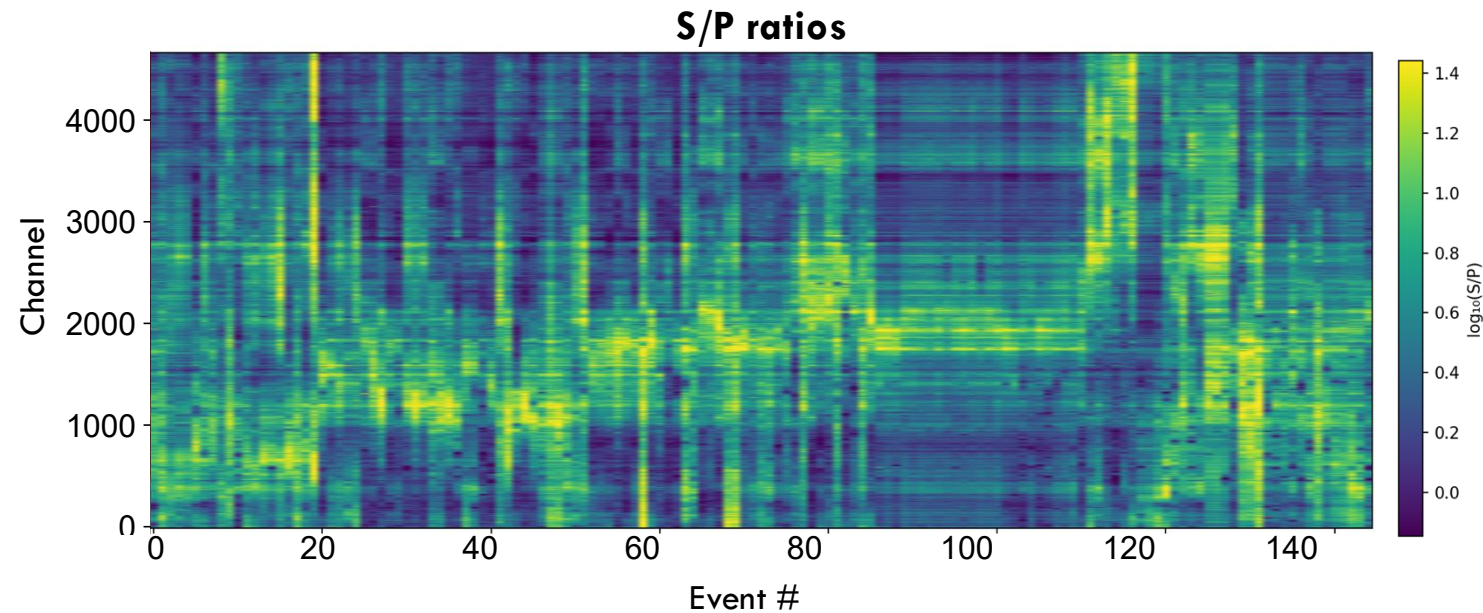
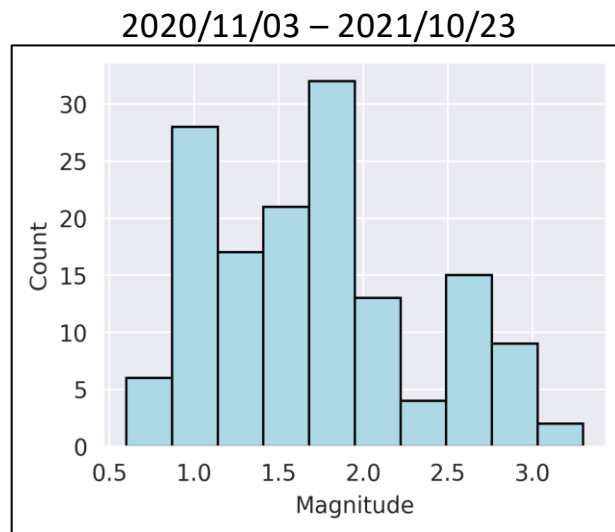
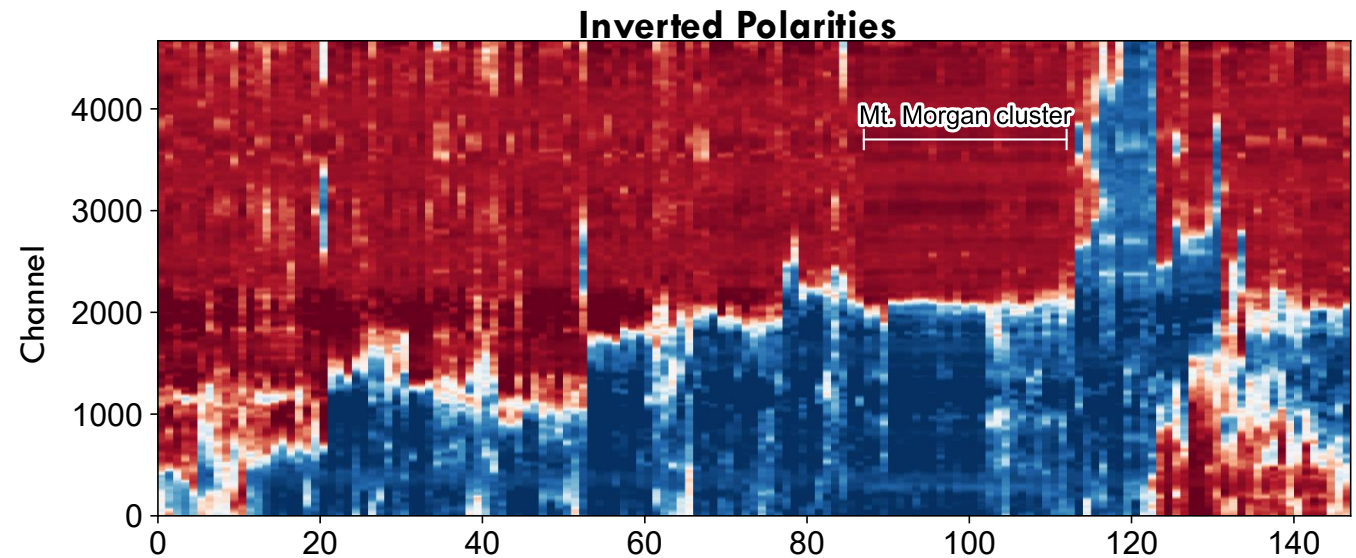
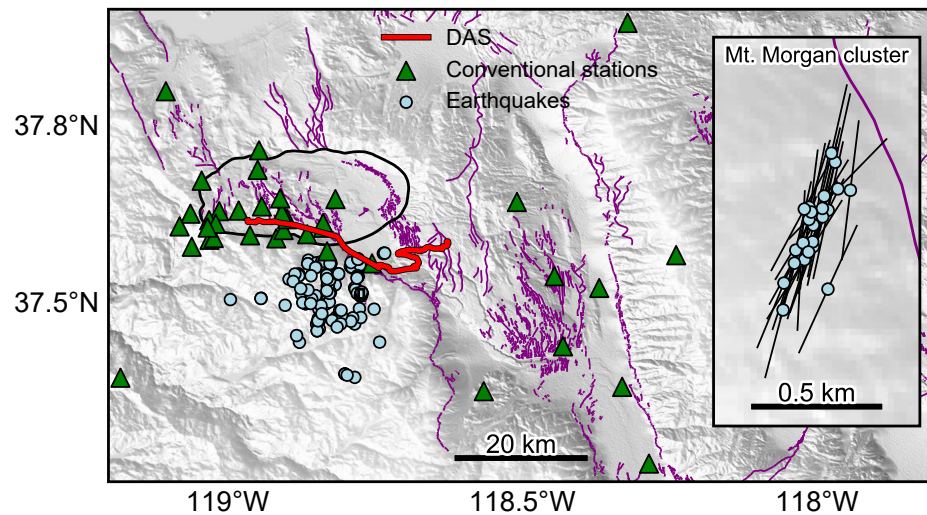
DASFM: workflow

- Notebook that runs the entire workflow:

[*lab3_simplest_running.ipynb*](#)



DAS polarities + S/P ratios

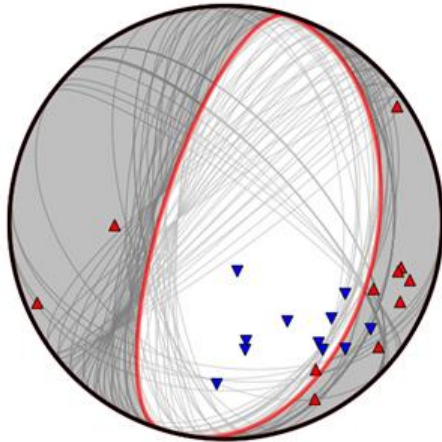


(Li et al., 2023)

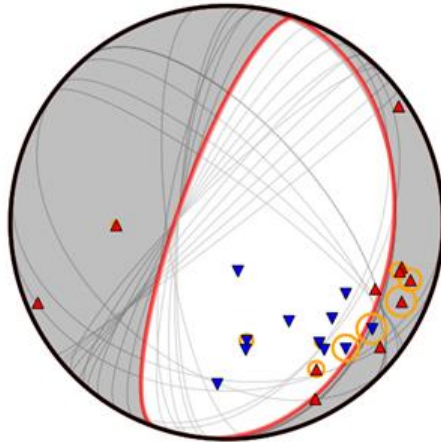
Inversion example

Event: nc73482301

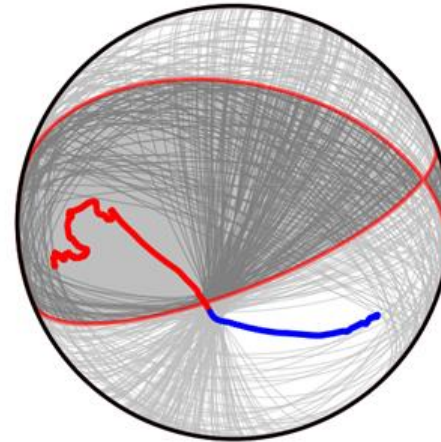
STA polarity (Quality: D)
prob:47% pol:13.6%
kagan:42.0 stdr:0.56



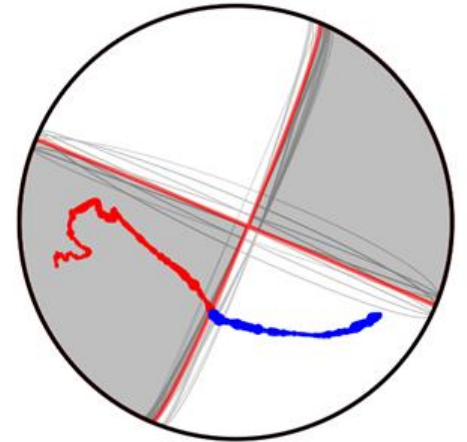
STA pol + STA S/P (Quality: D)
prob:60% pol:13.6%
kagan:43.4 stdr:0.56 sp:16.1%



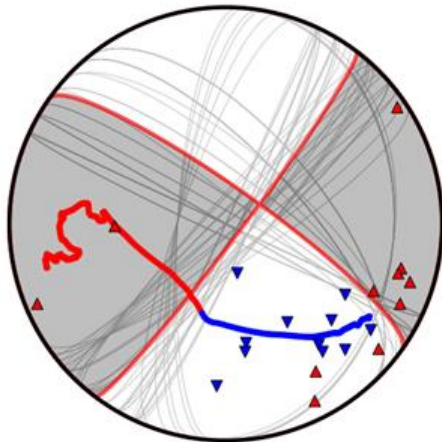
DAS polarity (Quality: D)
prob:21% pol:0.5%
kagan:59.5 stdr:0.69



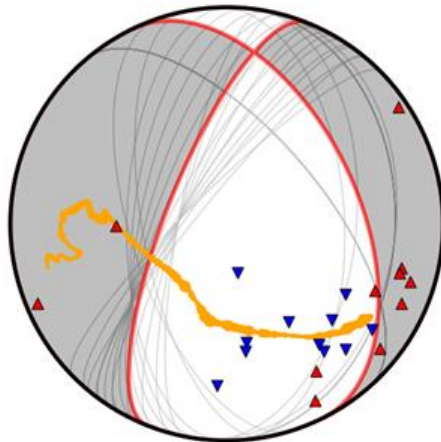
DAS pol + DAS S/P (Quality: A)
prob:100% pol:0.0%
kagan:7.6 stdr:0.69 sp:0.0%



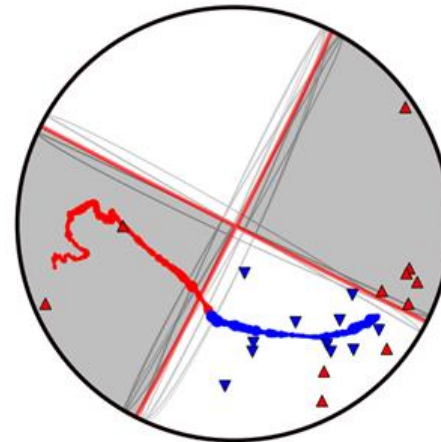
STA + DAS polarity (Quality: B)
prob:85% pol:8.0%
kagan:34.6 stdr:0.73



STA pol + DAS S/P (Quality: D)
prob:85% pol:18.2%
kagan:25.1 stdr:0.73 sp:18.9%



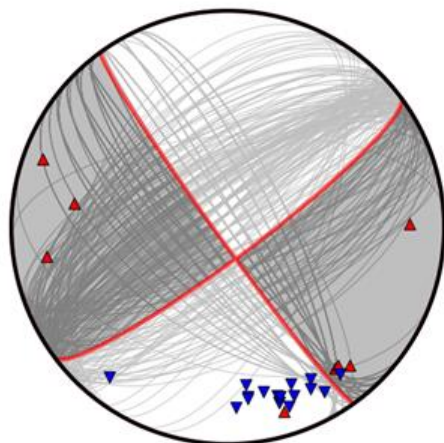
STA pol + DAS pol + DAS S/P (Quality: A)
prob:100% pol:7.9%
kagan:6.4 stdr:0.67 sp:0.2%



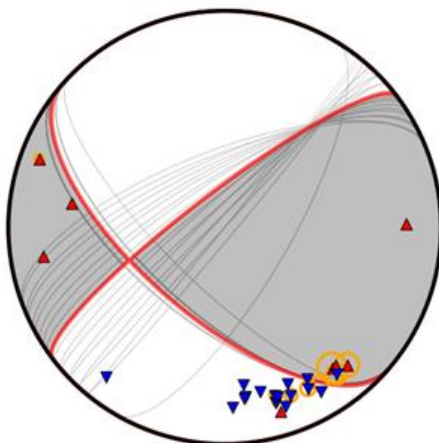
Inversion example

Event: nc73511090

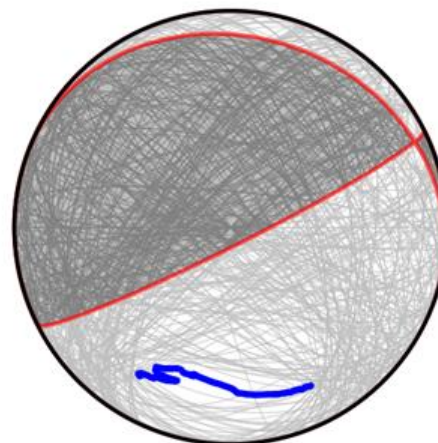
STA polarity (Quality: D)
prob:40% pol:8.0%
kagan:38.7 stdr:0.63



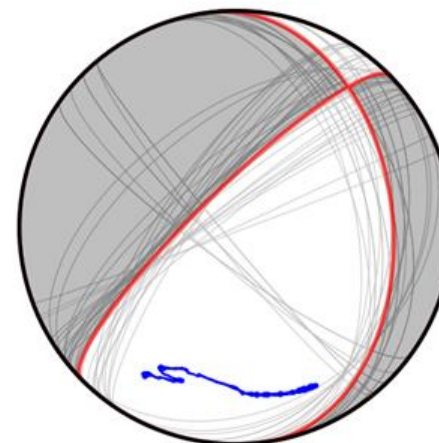
STA pol + STA S/P (Quality: A)
prob:95% pol:8.0%
kagan:17.4 stdr:0.54 sp:0.7%



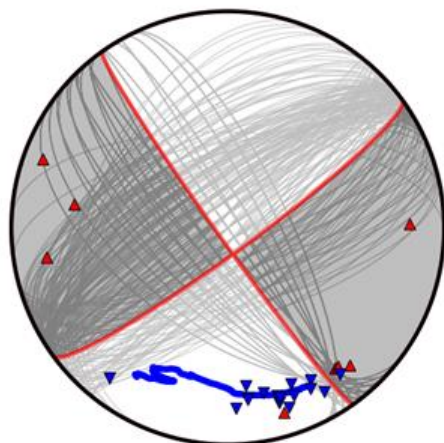
DAS polarity (Quality: D)
prob:12% pol:0.0%
kagan:61.6 stdr:0.84



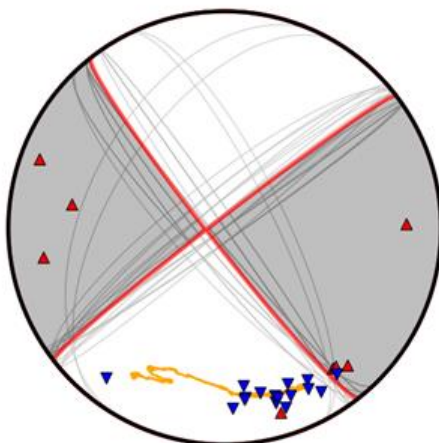
DAS pol + DAS S/P (Quality: C)
prob:76% pol:0.0%
kagan:42.6 stdr:0.72 sp:0.0%



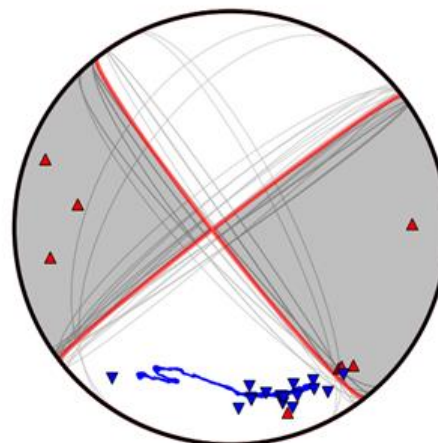
STA + DAS polarity (Quality: D)
prob:39% pol:4.0%
kagan:37.5 stdr:0.67



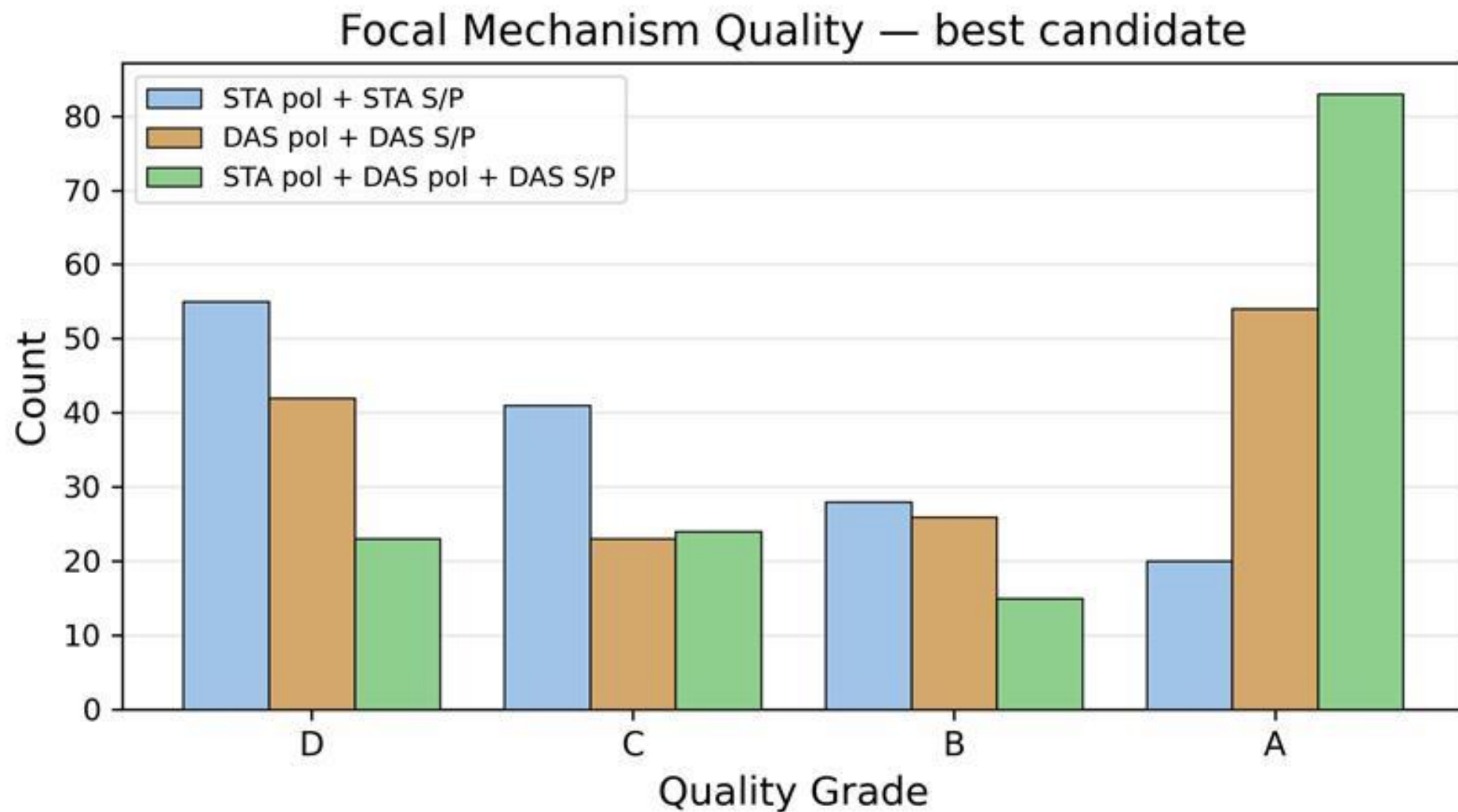
STA pol + DAS S/P (Quality: A)
prob:90% pol:8.0%
kagan:22.2 stdr:0.73 sp:0.2%



STA pol + DAS pol + DAS S/P (Quality: A)
prob:90% pol:4.0%
kagan:22.2 stdr:0.73 sp:0.2%



Inversion quality



Conclusions

- DAS earthquake recordings are dominated by near-surface scatterings.
- This enables us to, despite the single-component sensing on DAS:
 - ✓ Derive **P-wave vertical-displacement polarities** using cross-correlation of DAS P-wave windows.
 - ✓ Use DAS **S-P ratios** in a similar way to S-P ratios on conventional stations.
- DAS-based polarities and S-P ratios improve FM inversions.
- We developed a joint inversion framework that integrates DAS and conventional stations for FM inversions.

Acknowledgements

- We thank the California Broadband Cooperative for providing access to the Digital 395 telecommunication fibers.
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- We also thank the Doerr School of Sustainability Computational Team for their help in the GCP setup.
- We also thank SSA team for the opportunity to teach this workshop.



Thank you for your attention!
Questions?

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